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VERIFI

A Software Tool for Tracking Industrial Utility Data

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U.S. DEPARTMENT
of **ENERGY**

ORNL IS MANAGED BY UT-BATTELLE LLC
FOR THE US DEPARTMENT OF ENERGY

DOE ITO Software Tools help implement energy projects

50001 Ready Navigator

Energy management

- Resources
- Track progress



Understand
energy users

VERIFI



Facility energy use

- Utility bill tracking and analysis
- Utility/ GHG savings analysis

Understand
annual
energy use

MEASUR



System energy use

- Industrial System Analysis
- Treasure Hunts
- Inventory

JUSTIFI



Additional impacts

- Financial & performance metrics impacts of energy projects

Create and
promote energy
projects

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DOE ITO Software Tools are Free, Open, and Safe

VERIFI



The Utility Tracking Tool
Top-down Analysis

<https://verifi.ornl.gov/>

MEASUR



The Energy Assessment Tool
Bottom-up Analysis

<https://measur.ornl.gov>

JUSTIFI



The Project Valuation Tool
KPI Analysis

<https://justifi.ornl.gov>

Together, **MEASUR**, **VERIFI**, and **JUSTIFI** form a comprehensive suite of applications that help facilities measure energy performance, verify savings, and justify efficiency investments. **These tools include the following features:**

- Free and Easy to Use
- Private Data & Global Useability
- Generates Graphs, Figures, & Reports
- Cross-Platform & Open Source
- Accessible Help Text and Examples
- Regular Software Updates

Learn more at: <https://industrialresources.ornl.gov>

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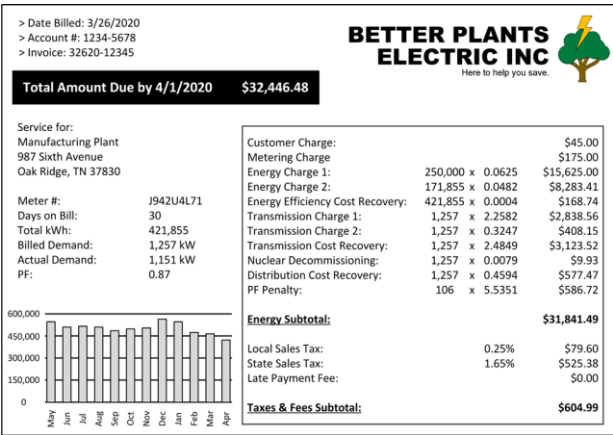
VERIFI - The Utility Tracking Tool

Facility personnel don't always have the time, resources, or knowledge to understand their energy spend: "it's **just another bill** I pay". Utility tracking services and software exist, but are **expensive**, especially for multi-site companies

VERIFI helps manufacturers track and understand their variable operating expenses **by themselves**.

Context

Who all is involved?
What is the scope?



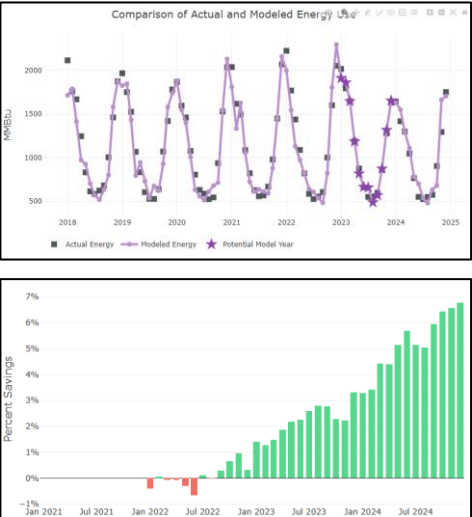
Understand

How do we collect, store, and analyze data?
What are my main energy users?



Continuous Improvement

What are our goals and how will we get there?
Training, Awareness, Documentation



	Baseline Year 2021	Report Year 2024
Number of Participating Plants*	5	5
Primary Energy Consumed (MMBtu)		
Electricity	506,192	483,553
Natural Gas	92,749	80,973
Total Primary Energy Consumed, (MMBtu)	598,942	564,526
Weather/Production/Other	6,579.9	
Normalizing related Adjustment for Baseline Primary Energy, (+/- MMBtu)		
Baseline Adjustment Due to Increase/Decrease in the Number of Facilities Reporting Relative to Baseline Year or Other Operational Changes, (+/- MMBtu)	74,702	
Adjusted Baseline of Primary Energy (MMBtu)	680,223	
New Energy Savings for Current Year (MMBtu)	20,593	
Unbanked Energy Savings (MMBtu)	115,697	
Total Energy Savings since Baseline Year (MMBtu)	115,697	
Annual Improvement In Energy Intensity for Current Year (%)**	3.45 %	
Total Improvement In Energy Intensity (%)**	17.01 %	

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Understand Facility-Level Energy Use

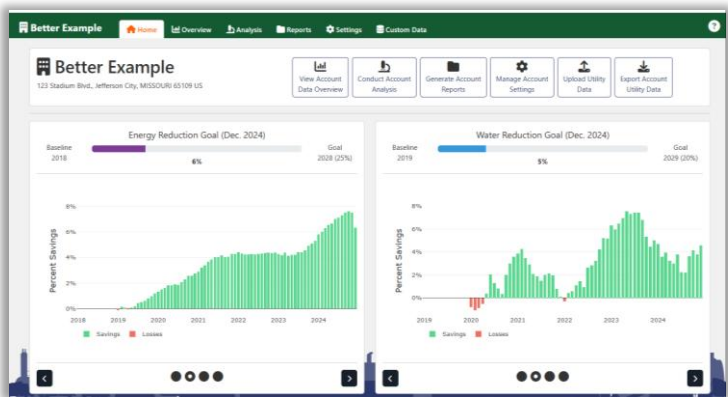
Helps you understand, visualize, and analyze utility bills and quantify utility reductions for multiple sites.

Key Features:

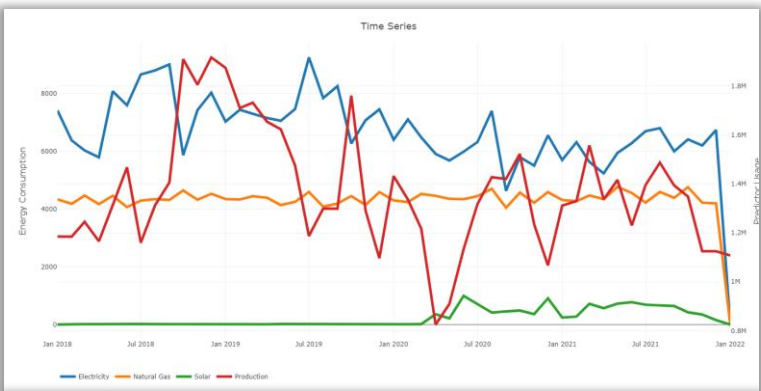
- Utility bill tracking, calendarization, visualization
- Energy, water, cost reduction analysis and reports
- Weather data for U.S. + more
- Accessible help text & examples + Regular updates

Additional Features:

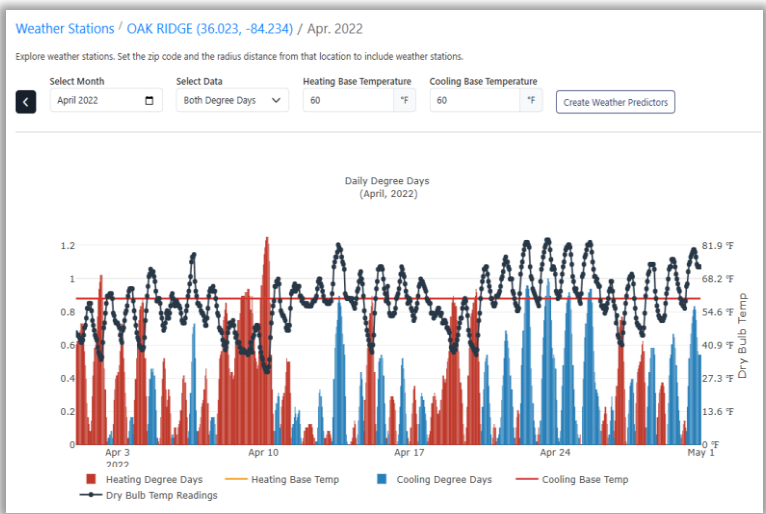
- FREE & Open source, Cross-platform tool
- Private Data & Global Useability



Find this tool at
<https://verifi.ornl.gov/>



	Energy (MMBtu)			Incremental Improvement	
Year	Actual	Modeled	Adjusted for Normalization	Total Savings % Improvement	Cummulative Savings
2018	32,240	32,302	32,240	—	—
2019	30,917	31,892	31,831	2.87 %	913.91
2020	19,407	21,232	21,171	8.33 %	2,677.7
2021	20,952	23,176	23,115	9.35 %	4,840



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Tool Highlights

- VERIFI is a company and facility-level energy analysis tool.
- It helps you track and visualize your utility bills.
 - Automatic calendarization
 - Annual costs, energy use, and water use
- Also helps energy management teams track and understand their utility and other outputs.
- Available for “Beta” testing
 - Replacement for legacy EnPI tool
 - Incorporate other DOE facility-level tools into a single program
 - Regular updates for development and to improve functionality
 - Updates will not affect data already in the tool

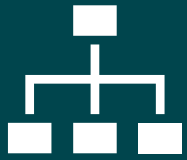


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Key Features

Tracking Utility Data



- Track energy usage, utility costs, and more
- Monitor facility and company level progress on key organizational goals

Analysis



- Automatic weather data based on facility location
- Regression modeling to understand energy intensity impact
- Roll up facility-level performance to company-wide performance

Data Visualization



- Easily identify unusual usage or high energy costs
- Determine correlation between energy use and relevant variables such as production or weather

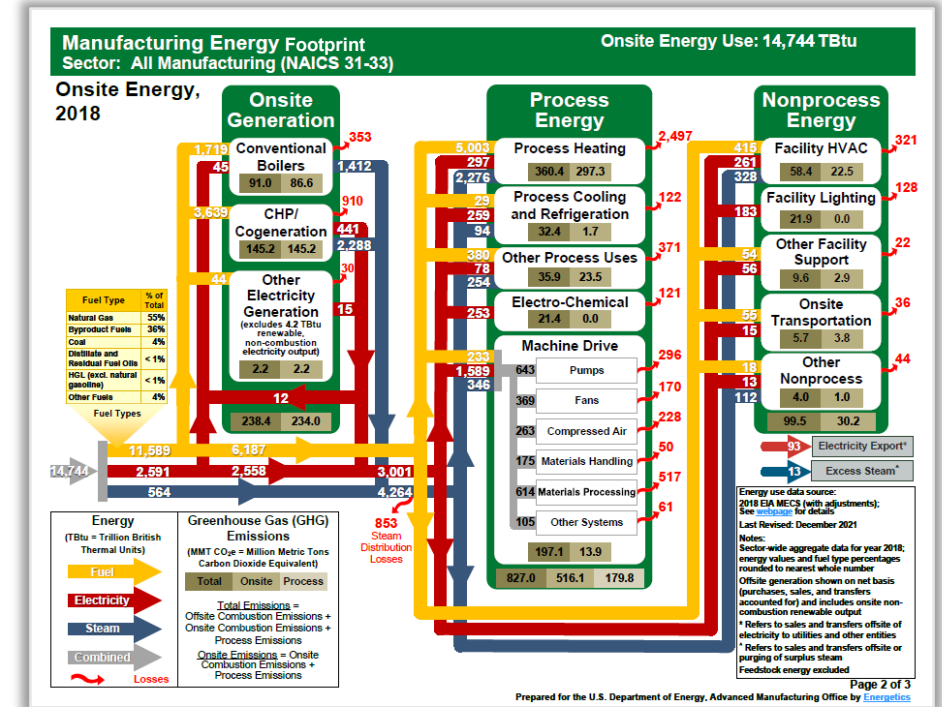
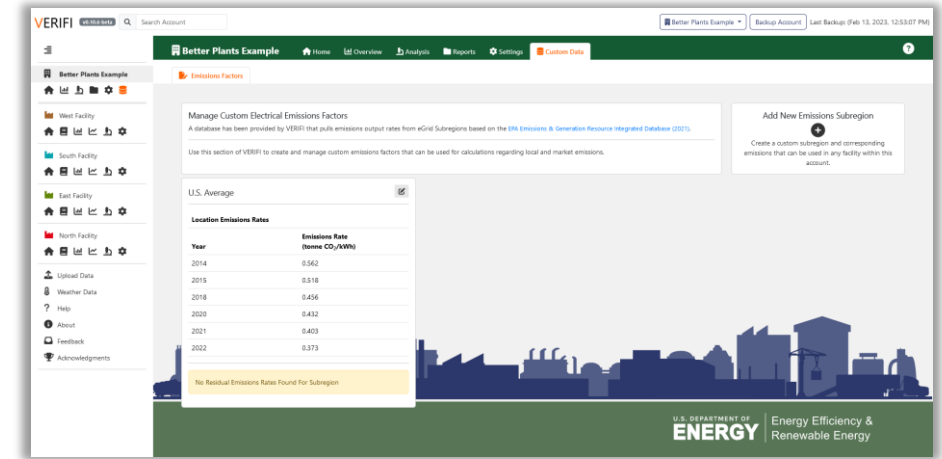
Reporting



- Automatically create annual reports for the Better Plants program
- Create facility comparison performance reports
- Keep track of your progress

How does VERIFI help me?

- Normalizes manufacturing energy use and generation.
- Helps identify energy losses and potential savings.
- Identifies high-performing and low-performing facilities with biggest impact on company-level performance.
- Provides simple analysis tools for absolute, intensity, and regression-based targets.
- Creates custom reports and visualizations.



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Creating Energy and Water Savings Analyses

1

Set Up Basic Info

Analysis Setup

Name: South Analysis

Baseline Year: 2018

Reporting Year: 2021

Energy Unit: Million British Thermal Units (MMBTU)

Fields set in "Settings" tab.

2

Choose Analysis Type

Natural Gas Setup

1. Select Analysis Type

- ☐ Absolute Energy Consumption
- ☐ Energy Intensity
- ☐ Modified Energy Intensity
- ☒ Regression

2. Add Baseline Adjustment

Has Baseline Adjustment: ☐

3. Regression Model

Use "Regression Model" tab to setup the regression model for this group.

3

Regression Made Easy!

Natural Gas Regression Model

Select Modeling Method: Calculate Models

Update Models

Current Models Generated: 1/10/23, 3:41 PM

Show Invalid Models: ☐

Regression Model

Analysis Variables

- CDD: ☒
- HDD: ☒
- Production: ☒

Use For Analysis

Regression Coefficient

Regression Constant: 102.12077904339644

Model Year: 2021

Additional Model Notes

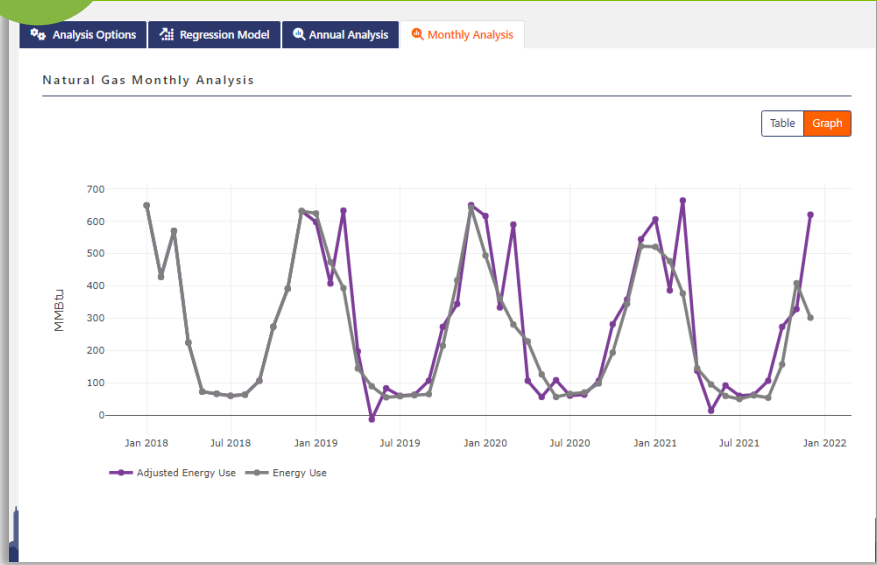
Select Model	Model Year	Variable p-Values	R2	Adjusted R2	Model p-Value	Formula	Model Notes
☒	2021	HDD: 0.00	0.582	0.540	0.00	102 + (0.429*HDD)	No production variable in model
☐	2021	Production: 0.01	0.553	0.508	0.01	880 + (-0.0219*Production)	Production coef < 0
☐	2018	HDD: 0.01 Production: 0.17	0.591	0.500	0.02	901 + (0.603*HDD) + (-0.0186*Production)	Production coef < 0 Adjusted R2 < .5
☐	2020	CDD: 0.02 Production: 0.09	0.583	0.490	0.02	160 + (-0.435*CDD) + (0.00550*Production)	CDD coef < 0 Adjusted R2 < .5

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Creating Energy and Water Savings Analyses

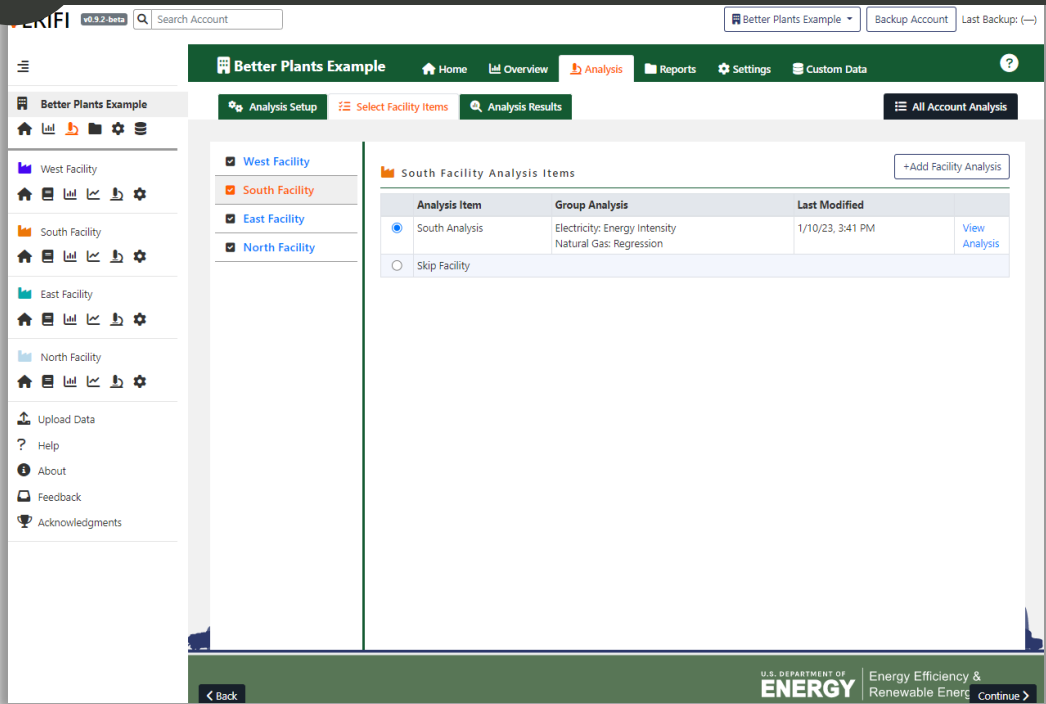
4

View Results



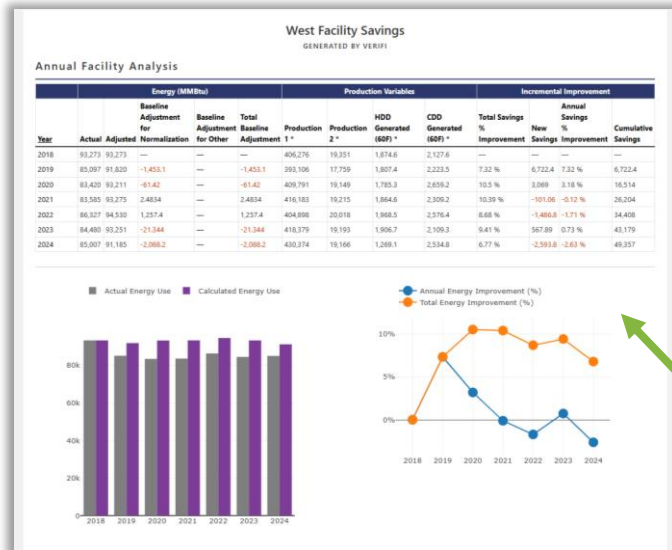
5

Combine Multiple Sites

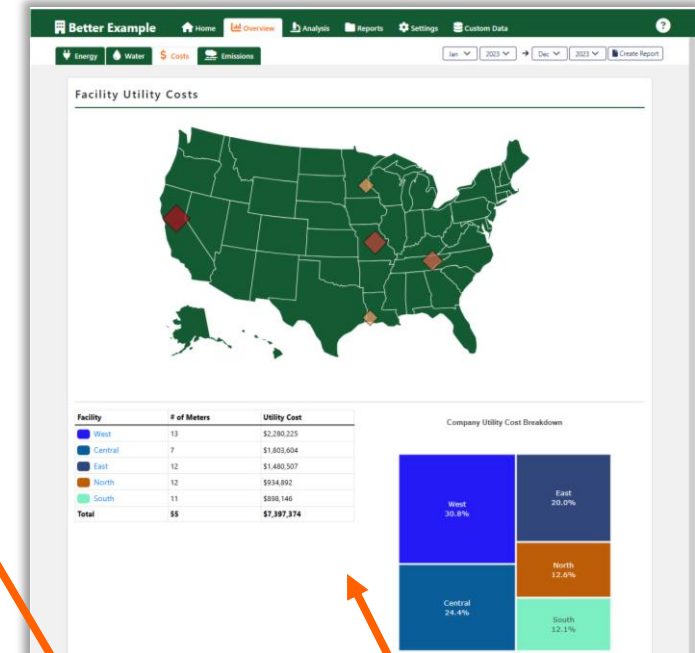


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Several **dashboards and reports** to help keep an eye on energy, other utility, cost reduction efforts.



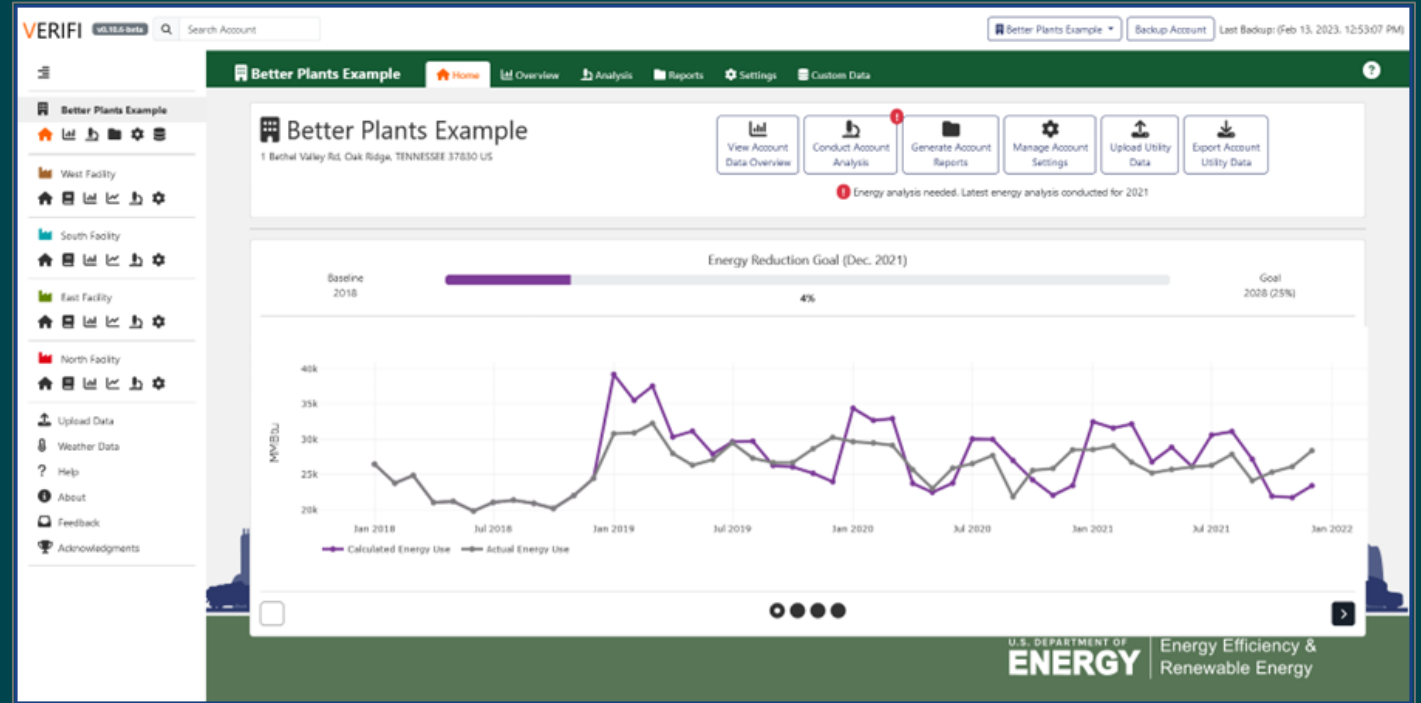
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Dashboards can show energy usage and costs.

Finding VERIFI

- Updates occur often
 - Automatic notification of updates if connected to the internet
 - Please report bugs to: verifi-help@ornl.gov
- Download from DOE or GitHub
 - Note: Uninstalling will delete your data
- If using online –
 - Note: Clearing cache will delete your data



Where to find this tool:

- Download Version: <https://industrialresources.ornl.gov>
- Web Version: <https://verifi.ornl.gov/>
- Google Web Search: "VERIFI"

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Setup Overview

Getting data into VERIFI and visualizing your usage.



First Time Starting VERIFI

VERIFI v0.14.2-beta Account Required ▾

Welcome to VERIFI

VERIFI is a tool used to track and analyze utility consumption data at both a corporate and facility level.

There are a few options for getting started.

Create New Account

Fill out the account, facility, meter, and predictor data details. Once data has been added, users can conduct an analysis on the performance of their facilities.

1 [+ Get Started](#)

Upload Account Backup

Use this method if you have an account backup file that was exported from a different VERIFI system.

2 [Upload Backup](#)

Load Example Account

Load in an example account to let you test out VERIFI's capabilities with example data created to mirror a real world company.

3 [Load Example](#)

4 About VERIFI and ORNL's Industrial Resources

For more information about VERIFI, as well as information about similar resources developed by ORNL, please visit [ORNL's Industrial Resources](#). Here you will find a collection of guidance and user support documents.

Dynamic help text is provided throughout VERIFI. If you need help at any time within the tool, click or hover the icon and help text will appear.

5 Contact and Feedback

If you have any feedback or would like to contact the team behind VERIFI, please reach out to us at verifi-help@ornl.gov.

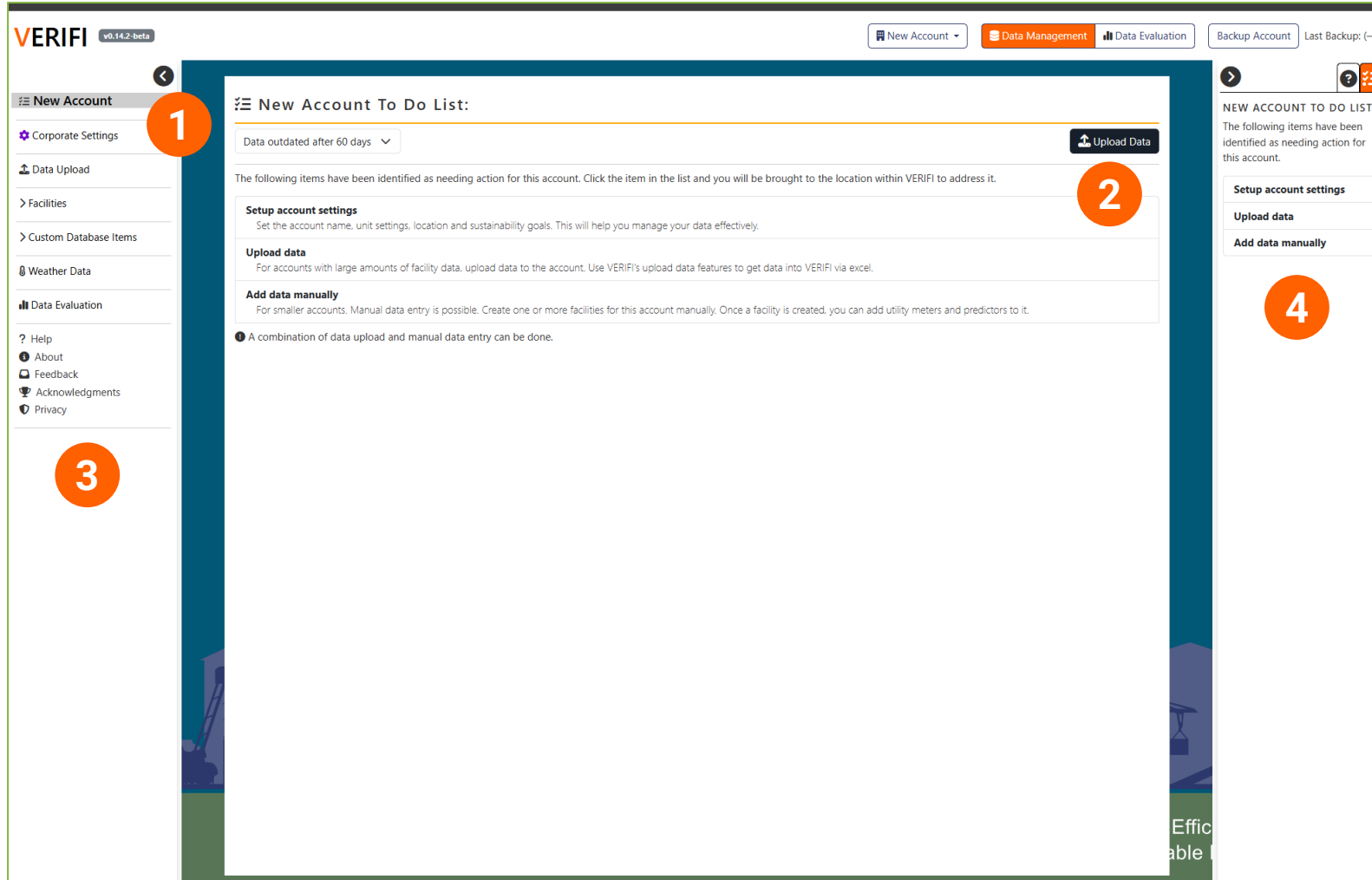
U.S. DEPARTMENT OF **ENERGY** | Energy Efficiency & Renewable Energy

1. Create Account
2. Load Backup data
3. Load sample data
4. Find more information about VERIFI and other Tools
5. Contact the team for help

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VERIFI starts you with a simple “To Do” list



1. Create the account

- Give company name and other details

2. Import Data

- Via spreadsheet
- Manually

3. Once you have entered data, navigate via the Left Panel

4. Revisit your “To Do” list in Data Management anytime via Right Panel or Account Name at top of Left Panel

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Account Setup

1. Enter Company Information: (Sets default for facilities)

- Name (Non-Optional)
- Point of contact (Optional)
- ZIP Code (useful for weather & GHGs)
- NAICS Code (Optional)

Corporate Information

Account Name *

New Account

Country

United States of America (the)

Address

[Search Address](#)

State

City

Zip

NAICS Code

Notes

Contact Name

Contact Phone

Contact Email

2. Enter Data Collection Defaults:

- Default units for different types of meters
- IPCC reporting preference & eGRID Subregion

Units of Measure

Custom

Boundary for analysis and display

☒ Source Energy

☐ Site Energy

Result Energy Unit

Million British Thermal Units (MI)

Default units for setting up meters

Electricity Unit

Kilowatt-hour (kWh)

Volume Unit (Liquid)

Gallon (gal)

Volume Unit (Gases)

Standard Cubic Feet (SCF)

Mass Unit

Pounds (lb)

Electricity Global Warming Potential

IPCC Assessment Report Version

☐ AR4

☒ AR5

eGRID Subregion

SRTV

[View Emission Rates](#)

3. Enter Goals Information:

- Default units for different types of meters
- IPCC reporting preference & eGRID Subregion

Sustainability Questions

Is this account a member of the Better Plants Program?

☐ Better Plants Program Partner

☒ 1. Do you have Energy Reduction goal?

Reduction Goal

25

%

Is goal for absolute energy or energy intensity?

☐ Absolute

☒ Intensity

Baseline Year

2025

Target Year

2035

☐ 2. Do you have a Greenhouse Emission/Carbon Reduction goal?

☐ 3. Do you have Renewable Energy goal?

☐ 4. Do you have a Waste Reduction goal?

☐ 5. Do you have a Water Reduction goal?

Financial Reporting

When is your fiscal year?

☒ Calendar Year

☐ Non Calendar Year

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Account Setup

Automatic Account Backup

- Setup a secondary file location where your data is saved every time you open the app (**only available on download version**)

Import Backup Data

- Import a VERIFI json file

Connect Utility Bills

- Select a folder where you can store utility bills to connect them to VERIFI bill entries (**only available on download version**)

Backup Account

- Manually create a backup file

Delete Account

- Remove this account file from your VERIFI

 Automatic Account Backup

Save an automatic backup of this account's data to your machine or a shared drive. The backup can be used to restore the data if it is lost or if you want to revert back to an older state of the account.

Select Automatic Backup Location

Import Backup Data

Use the below button to import backup data files. Accounts or facilities can be imported as new data or to overwrite existing data.

Import Backup File

Connect Utility Bills

Select a folder on your machine where you want to save your utility bills. All the bills will be copied to the selected folder.

Select Folder

Backup Account

You can protect your files from accidental loss by creating a backup. We recommend backing up your data frequently. This button will download a copy of the selected account data. If you want just facility information then selected the facility from the table above and export from that page.

Backup Account

☐ Download As .zip File

Delete Account

Once deleted, you will permanently lose all your data linked to your account. Please make a backup of your data before proceeding with this step.

Delete Account



Facility Setup

- **Very similar to company setup**
 - Inherits traits from company setup
- **Each facility can have different settings for:**
 - Points of contact
 - Collection units
 - Reporting goals
- **Color code facilities for easy navigation**
- **Manage Facilities: Create facilities & arrange for all sites in your portfolio**

Facility Information

Facility Name *
New Facility

Country
United States of America (the)

Address [Search Address](#)

State
City

Zip
37830

NAICS Code
Facility Classification
Manufacturing

Facility Area
ft²
Facility Color

Notes

Contact Name
Contact Phone

Contact Email

Facility Units

Units of Measure
Custom

Boundary for analysis and display
☒ Source Energy
☐ Site Energy

Result Energy Unit
Million British Thermal Units (MI)

Default units for setting up meters

Electricity Unit
Kilowatt-hour (kWh)

Volume Unit (Liquid)
Gallon (gal)

Volume Unit (Gases)
Standard Cubic Feet (SCF)

Mass Unit
Pounds (lb)

Electricity Global Warming Potential

eGRID Subregion
SRTV

[View Emission Rates](#)

The eGRID Subregion and emissions output rate will be set based on your zipcode and the data found in the [EPA GHG Emissions Factors Hub](#).
Custom emissions rates can be added and managed in the [Account Custom Data](#) page.

Account Facilities

Upload Data

Add Facilities

	Facility	State	Country	Actions
▼	West	CALIFORNIA	US	
▼	South	LOUISIANA	US	
▼	East	TENNESSEE	US	
▼	North	MINNESOTA	US	
▼	Central	MISSOURI	US	

Basic VERIFI Layout – Data Management

- **Data Management**

- Upload data, change meter settings, include weather data, review data (data quality & charges)

Left Panel

- View / switch between all your facilities
- Review meters & predictors
- Data upload, weather data, help, feedback, acknowledgements

The screenshot displays the VERIFI Data Management interface. The top navigation bar includes the VERIFI logo, version 14.2-beta, and tabs for Better Example, Data Management (active), Data Evaluation, Backup Account, and Last Backup. The left sidebar shows a tree view with categories like Corporate Settings, Data Upload, Facilities (West, South, East, North, Central), Facility Settings, Meters (Manage Meters, Meter Grouping, Electricity, Municipal Water, Natural Gas, RECs, VPPA, Meter Settings, Meter Readings, Meter Data, Meter Monitoring), Predictors, and Product A. The main content area is titled 'Central Meters' and features a table with columns: Name, Source, Fuel, Scope, Collection Units, and Actions. The table lists five entries: VPPA, RECs, Natural Gas, Municipal Water, and Electricity. Each entry has corresponding data for Source, Fuel, Scope, and Collection Units, and an Actions column with icons for edit, delete, and add. A 'Copy Table' button is located below the table. The right sidebar shows a 'BETTER EXAMPLE TO DO LIST' section with a message: 'No items to do at this time. Proceed to data evaluation and analysis!'.

Name	Source	Fuel	Scope	Collection Units	Actions
VPPA	Electricity	—	Purchased Electricity	MWh	[Edit] [Delete] [Add]
RECs	Electricity	—	Purchased Electricity	MWh	[Edit] [Delete] [Add]
Natural Gas	Natural Gas	—	Stationary	MMBtu	[Edit] [Delete] [Add]
Municipal Water	Water Intake	—	—	kgal	[Edit] [Delete] [Add]
Electricity	Electricity	—	Purchased Electricity	kWh	[Edit] [Delete] [Add]

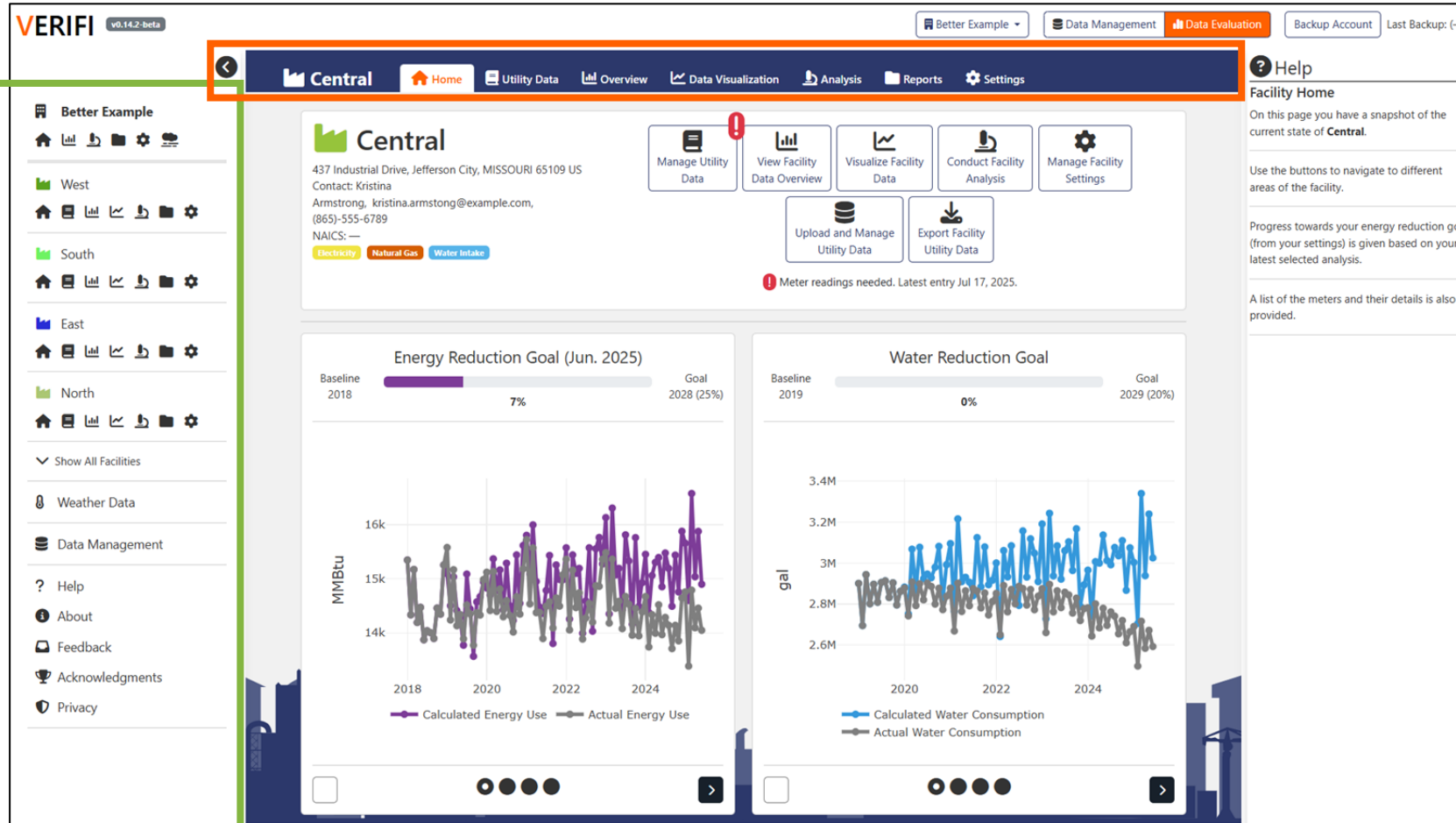
Right Panel

- Review To Do List
- Find help text for each page of the tool

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Basic VERIFI Layout – Data Evaluation

- Data Evaluation
 - Review data, visualize data, conduct analyses, create reports



Side Panel

- View all your facilities
- Switch between home, utility data, data visualization, analysis, and settings views
- Data upload, weather data, help, feedback, acknowledgements

Top Panel

- Use tabs to navigate steps and menus
- Find help text for each page of the tool

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Navigating with the icons



Home

View progress towards goals



Utility Data

Enter and view data



Overview

High level view without analysis



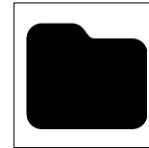
Data Visualization

Create figures from your data



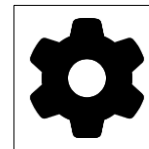
Analysis

Perform different analyses (regression) on data



Reports

Create reports from analyses



Settings

Revisit the setup pages

Importing Data into VERIFI

VERIFI v0.14.2-beta

Better Example ▾ Data Management Data Evaluation Backup Account Last Backup: (—)

BETTER EXAMPLE

- Corporate Settings
- Data Upload**
- Facilities
 - Manage Facilities
 - West
 - South
 - East
 - North
 - Central
- Custom Database Items
- Weather Data
- Data Evaluation
- Help
 - About
 - Feedback
 - Acknowledgments
 - Privacy

Upload Data From .XLSX Files

Data Entry Options

VERIFI has two options for adding utility data to corporate accounts. Users with only a couple facilities may want to add data manually by filling out data forms for facilities, meters, bills and predictors. Otherwise, accounts with many facilities can use our excel template for uploading all the data tracking information at once.

It should be noted that a combination of manual data entry and excel data uploading can be used.

An excel template has been created to more easily upload facility and utility data. [Download Template](#)

Choose Files No file chosen

- Select the files you wish to upload data from into facilities.
- Data must be in excel (.XLSX) files.
- Once the files have been selected we will work through parsing out the relevant data into the correct places.
- Click the "Choose Files" button or drag and drop files into the grey box.
- Multiple files can be uploaded.

BETTER EXAMPLE TO DO LIST:

The following items have been identified as needing action for this account.

No items to do at this time. Proceed to data evaluation and analysis!

Upload files to get started!

In Data Management

1. Select "Upload Data" from side panel
2. Select "Choose File" to browse for upload file or drag and drop.
3. Follow prompts from VERIFI to complete the data upload.

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Importing Data into VERIFI – Excel Template

- Use the VERIFI Excel template to set up multiple facilities, meters, years of data entry...
All at once!
- Select “Upload Data” and then download blank template
- Follow tabs to setup meters and collect billing and predictor data

Upload Data From .XLSX Files

Data Entry Options

VERIFI has two options for adding utility data to corporate accounts. Users with only a couple facilities may want to add data manually by filling out data forms for facilities, meters, bills and predictors. Otherwise, accounts with many facilities can use our excel template for uploading all the data tracking information at once.

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Choose Files

No file chosen

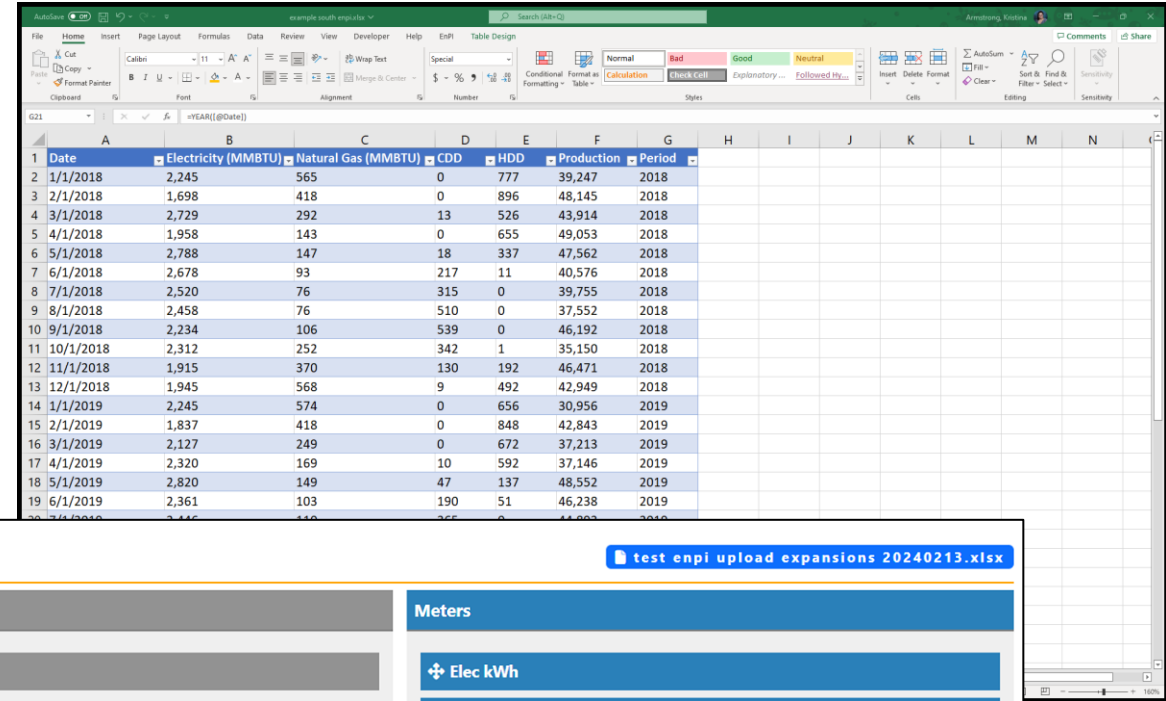
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- Once the files have been selected we will work through parsing out the relevant data into the correct places.
- Click the “Choose Files” button or drag and drop files into the grey box.
- Multiple files can be uploaded.

✓ Facility Setup

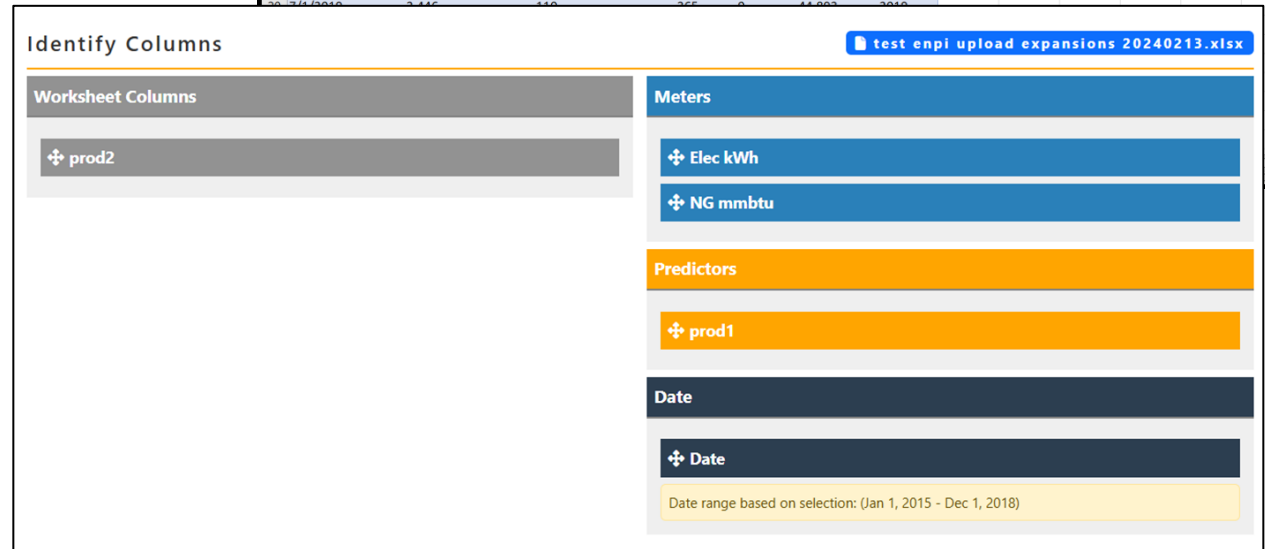
Facility Name	Address	Country	U.S. State	City	ZIP Code
West	1845 Scott St.	US	CALIFORNIA	Sacramento	94203
South	1234 Anderson Lane	US	LOUISIANA	Lake Charles	70601
East	1 Bethel Valley Rd	US	TENNESSEE	Oak Ridge	37830
North	2323 Orange Elm Court	US	MINNESOTA	Minneapolis	55111
Central	437 Industrial Drive	US	MISSOURI	Jefferson City	65109

Importing Data into VERIFI – Data Wizard

- Use the data import wizard to enter legacy EnPI files or similar column formatted data
- VERIFI still walks you through the process
- Assign columns a data type
 - Date, meter, predictor
- Assign meters to facilities
- Assign predictors to facilities
- Repeat this process for all facilities in portfolio



Date	Electricity (MMBTU)	Natural Gas (MMBTU)	CDD	HDD	Production	Period
1/1/2018	2,245	565	0	777	39,247	2018
2/1/2018	1,698	418	0	896	48,145	2018
3/1/2018	2,729	292	13	526	43,914	2018
4/1/2018	1,958	143	0	655	49,053	2018
5/1/2018	2,788	147	18	337	47,562	2018
6/1/2018	2,678	93	217	11	40,576	2018
7/1/2018	2,520	76	315	0	39,755	2018
8/1/2018	2,458	76	510	0	37,552	2018
9/1/2018	2,234	106	539	0	46,192	2018
10/1/2018	2,312	252	342	1	35,150	2018
11/1/2018	1,915	370	130	192	46,471	2018
12/1/2018	1,945	568	9	492	42,949	2018
1/1/2019	2,245	574	0	656	30,956	2019
2/1/2019	1,837	418	0	848	42,843	2019
3/1/2019	2,127	249	0	672	37,213	2019
4/1/2019	2,320	169	10	592	37,146	2019
5/1/2019	2,820	149	47	137	48,552	2019
6/1/2019	2,361	103	190	51	46,238	2019



Identify Columns

test enpi upload expansions 20240213.xlsx

Worksheet Columns

- prod2

Meters

- Elec kWh
- NG mmbtu

Predictors

- prod1

Date

- Date

Date range based on selection: (Jan 1, 2015 - Dec 1, 2018)

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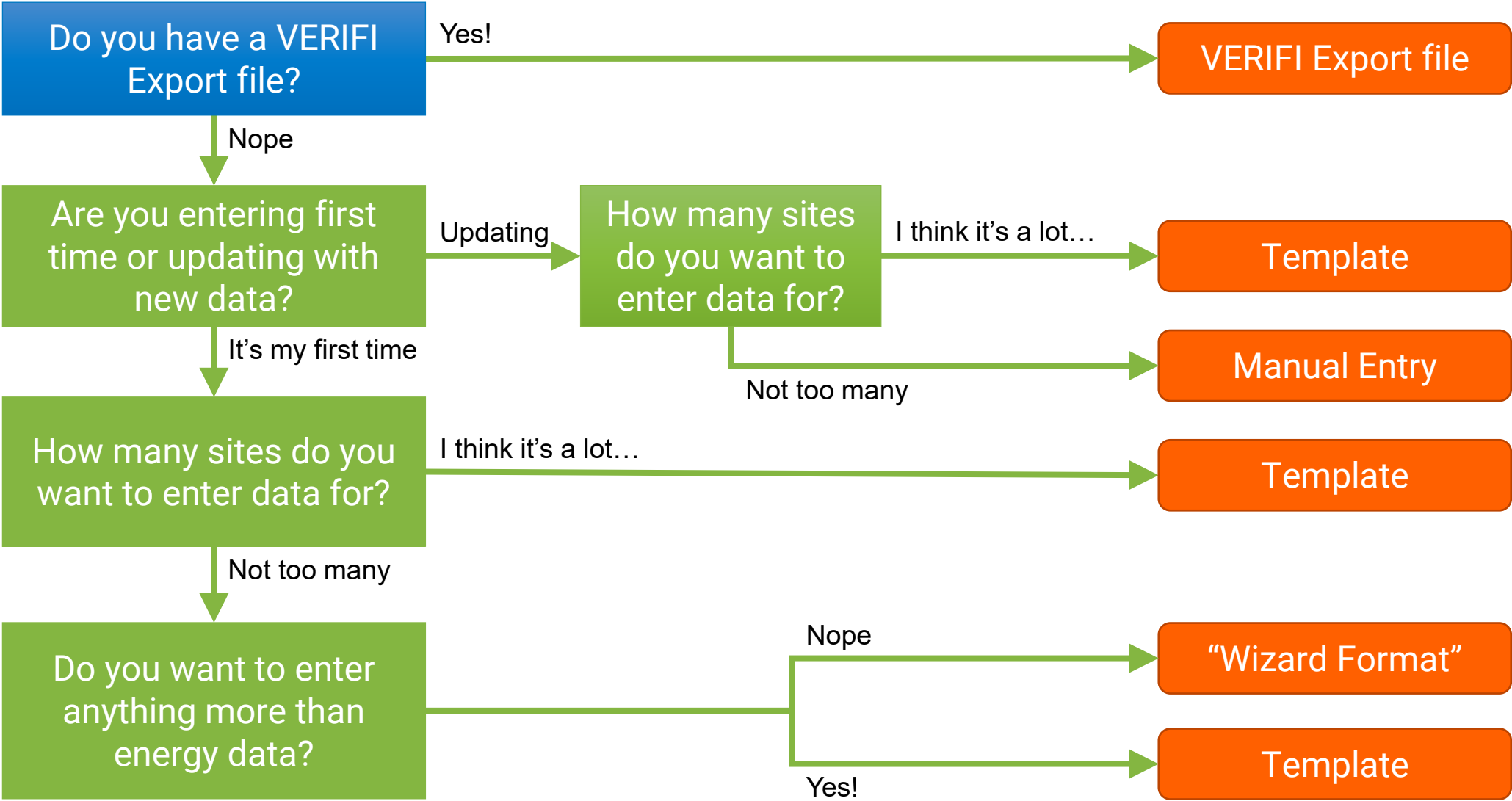
Importing Data into VERIFI – Manual Entry

- Create meters individually
- For each meter, add meter readings
 - Includes meter read date
 - Consumption/ loss
 - Costs
- Works best for monthly updating, rather than entering historic data

The image displays two screenshots of the VERIFI v0.14.2-beta web application interface. The top screenshot shows the 'New Meter' form, which includes sections for 'Meter Information' (Name, Source, Collection Unit, Site to source), 'Other Information' (Meter Number, Account Number, Building / Location, Utility Supplier, Notes), and 'Meter Charges' (Add Charge, Contract Agreement). The bottom screenshot shows the 'Electricity Meter Readings' form, which includes sections for 'General Information' (Meter Read Date, Is Estimated Reading, Total Real Demand, Total Energy Use, Total Market-Based Emissions, Total Location-Based Emissions), 'Detailed Charges' (Add charges), and a summary section (Total Billed Demand, Power Factor, Total Cost). Both screenshots show a sidebar with navigation options like 'Facilities', 'Meters', and 'New Meter'.

DRAFT

Importing Data into VERIFI – Manual Entry



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Importing Data into VERIFI – Manual Entry

- In VERIFI – all data is associated with a “meter”
- For utilities, this can represent individual billed meters or pre-grouped data points for each utility.
- For non-utilities (mobile fuel use, other outputs, and green financial agreements), this represents the amount used, purchased, or released over a period.
 - You can enter data quarterly or annually and select “Annualize over year” for Calendarization
- You can combine meters into a group for analysis
- Starting with individual meters allows users to:
 - Monitor areas of their facility
 - Group meters in different ways to explore how predictors impact utility use
 - Better calendarize data if meters are read on different days



DRAFT

Quickly view meter data

Meter Readings

Raw data exactly as entered: meter read date, charges, bill units

Better Example

Corporate Settings

Data Upload

Facilities

- Manage Facilities
 - West
 - South
 - East
 - North
 - Central
- Facility Settings
 - Meters
 - Manage Meters
 - Meter Grouping
 - Electricity
 - Meter Settings
 - Meter Readings
 - Meter Data Quality Re...
 - Meter Monthly Data
 - Municipal Water
 - Natural Gas
 - RECs
 - VPPA

Electricity Meter Readings

Choose Columns Data Quality Report 12 Import Data + Add New Bill

General Information					Emissions				
Meter Read Date	Total Energy (kWh)	Total Cost	Total Real Demand (kW)	Total Billied Demand (kW)	Power Factor	Total Market-Based Emissions (tonne CO ₂ e)	Total Location-Based Emissions (tonne CO ₂ e)	RECs (MWh)	Actions
☐ Jul 17, 2025	1,270,752	\$132,900	1,764	1,764		812.06	718.8	0	
☐ Jun 17, 2025	1,298,695	\$135,211	1,745	1,745		829.91	734.6	0	
☐ May 17, 2025	1,206,733	\$126,844	1,733	1,744		771.15	682.59	0	
☐ Apr 18, 2025	13,756,300	\$142,081	1,736	1,749		8,790.8	7,781.2	0	
☐ Mar 16, 2025	1,088,831	\$116,329	1,744	1,749		695.8	615.9	0	
☐ Feb 18, 2025	1,247,085	\$130,057	1,676	1,749		796.93	705.41	0	
☐ Jan 18, 2025	1,222,110	\$127,557	1,642	1,749		780.97	691.28	0	
☐ Dec 18, 2024	1,288,499	\$133,796	1,677	1,749		823.4	806.51	0	
☐ Nov 16, 2024	1,195,169	\$125,263	1,659	1,749		763.76	748.09	0	
☐ Oct 17, 2024	1,177,723	\$123,934	1,692	1,749		752.61	737.17	0	
☐ Sep 18, 2024	1,352,476	\$139,778	1,707	1,749		864.28	846.56	0	
☐ Aug 16, 2024	1,265,932	\$131,943	1,701	1,749		808.98	792.39	0	

Meter Monthly Data

Processed data: calendarized, converted to facility units

Better Example

Corporate Settings

Data Upload

Facilities

- Manage Facilities
 - West
 - South
 - East
 - North
 - Central
- Facility Settings
 - Meters
 - Manage Meters
 - Meter Grouping
 - Electricity
 - Meter Settings
 - Meter Readings
 - Meter Data Quality Re...
 - Meter Monthly Data
 - Municipal Water
 - Natural Gas
 - RECs

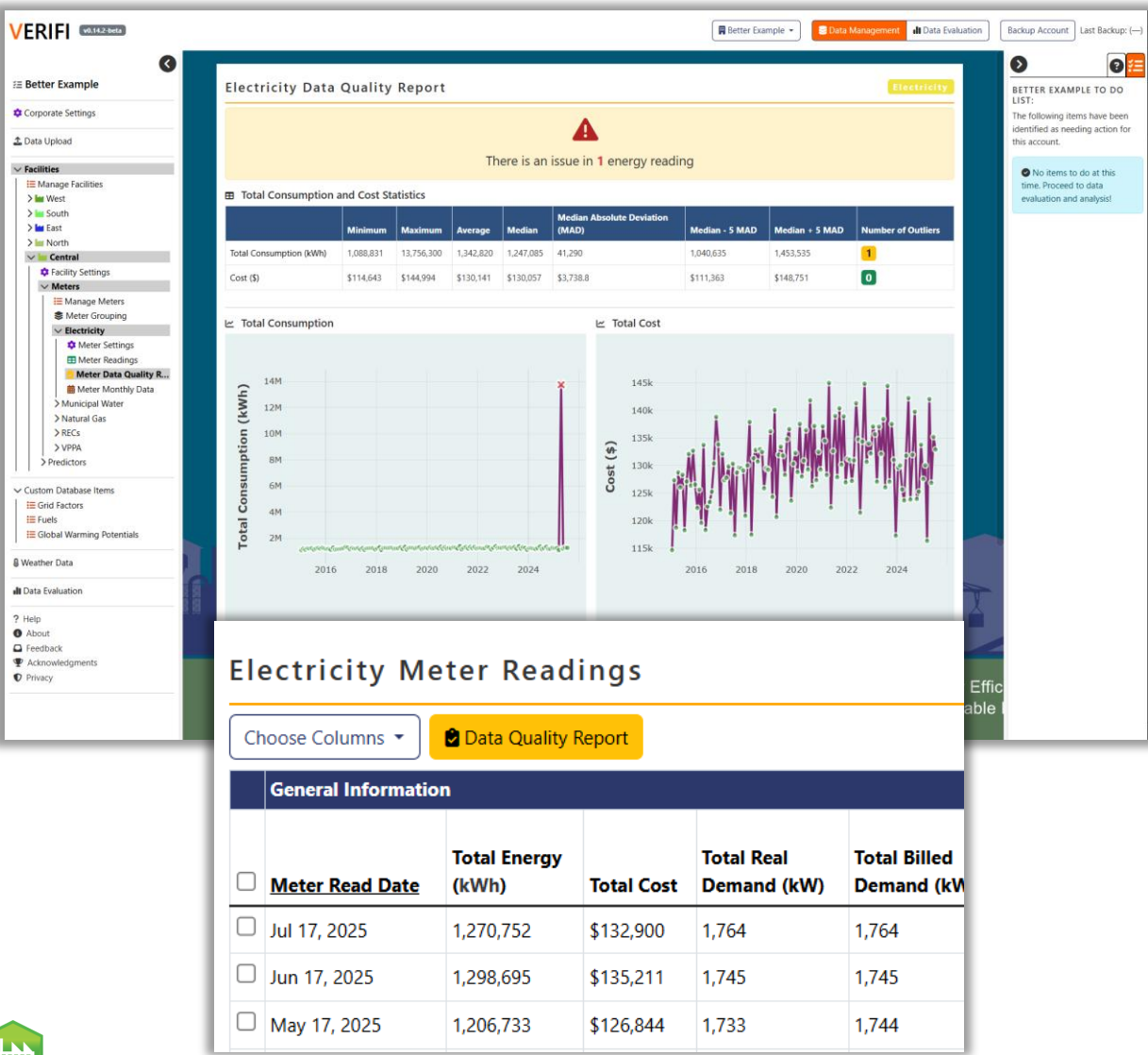
Electricity Calanderized Data

Filter Select Method 12 Table Graph Source: Energy Site Energy

Month	Total Energy (MMBtu)	Total Market-Based Emissions (tonne CO ₂ e)	Total Location-Based Emissions (tonne CO ₂ e)	Total Cost
June, 2025	12,927	807.01	714.33	\$131,738
May, 2025	13,245	826.85	731.89	\$135,420
April, 2025	81,919	5,114.1	4,526.8	\$129,986
March, 2025	70,866	4,424	3,915.9	\$136,170
February, 2025	11,699	730.35	646.48	\$120,259
January, 2025	12,617	787.67	697.21	\$128,606
December, 2024	12,665	790.67	774.45	\$128,752
November, 2024	12,295	767.57	751.83	\$125,342
October, 2024	12,776	797.6	781.25	\$131,107
September, 2024	12,540	782.85	766.8	\$127,526
August, 2024	12,981	810.39	793.77	\$131,635
July, 2024	13,124	819.32	802.52	\$134,603

Copy Table 1 2 3 4 5 11

Data quality reports



Quickly find data errors for meters and predictors

- Review data in graphs
 - Line with time and histograms
- Calculated outliers
 - Uses median instead of mean (MAD)
- Only informative, does not stop analysis
- Quickly see data quality evaluation in meter readings table
 - Green = All good
 - Yellow = Potential Errors

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Adding Utility Bill Charges

Add custom charges to bill readings

- Specify if it is related to consumption or demand, a flat charge, tax, or other kind of charge.

View relationships between entered charges and other data to understand what a charge is for.

Electricity

Electricity

Save Changes

Meter Information

Other Information

Name *

Electricity

Source *

Electricity

Collection Unit *

Kilowatt-hour (kWh)

Site to source *

3

Scope

Scope 2: Purchased Electricity

Demand Unit

Kilowatt (kW)

Meter Number

Central Electricity

Account Number

Building / Location

Utility Supplier

Notes

Meter Charges

Cancel

Save Changes

Name

Block 1

Block 2

General Information

Detailed Charges

Meter Read Date *

07/17/2025

Is Estimated Reading?

No

Total Real Demand

1764

kW

Total Energy Use *

1270752

kWh

Total Billed Demand

1764

kW

Total Market-Based Emissions

812 tonne CO₂e

Total Location-Based Emissions

719 tonne CO₂e

Power Factor

Total Cost

132899.56

\$

Block 1

0

kW

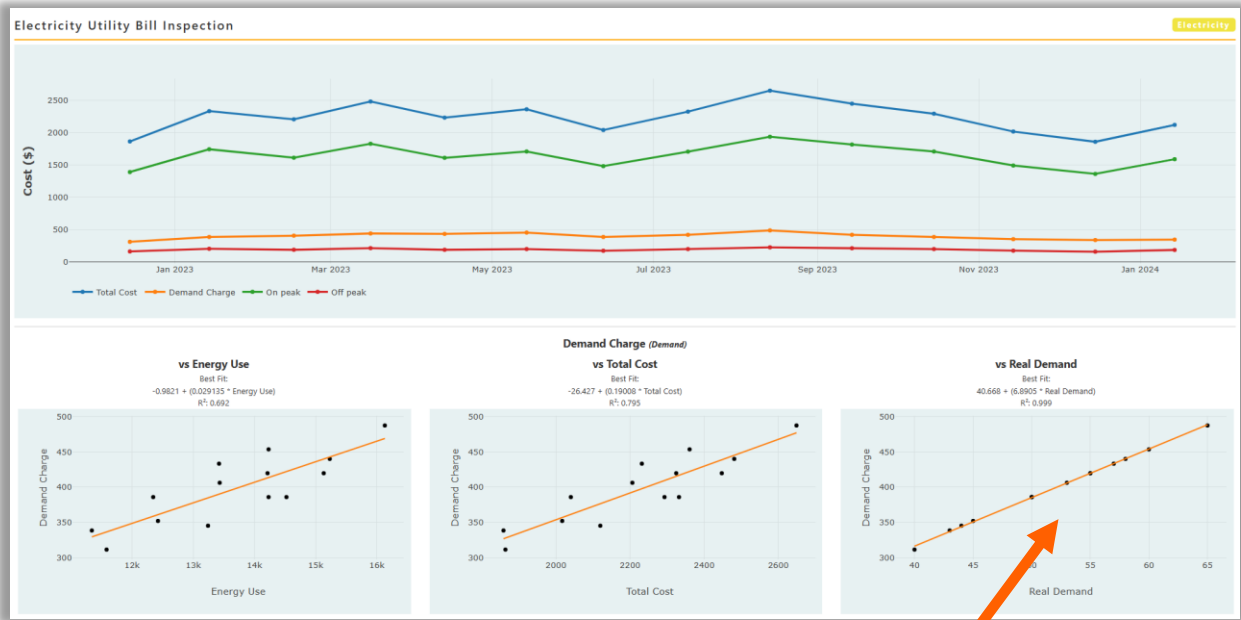
\$

Block 2

1

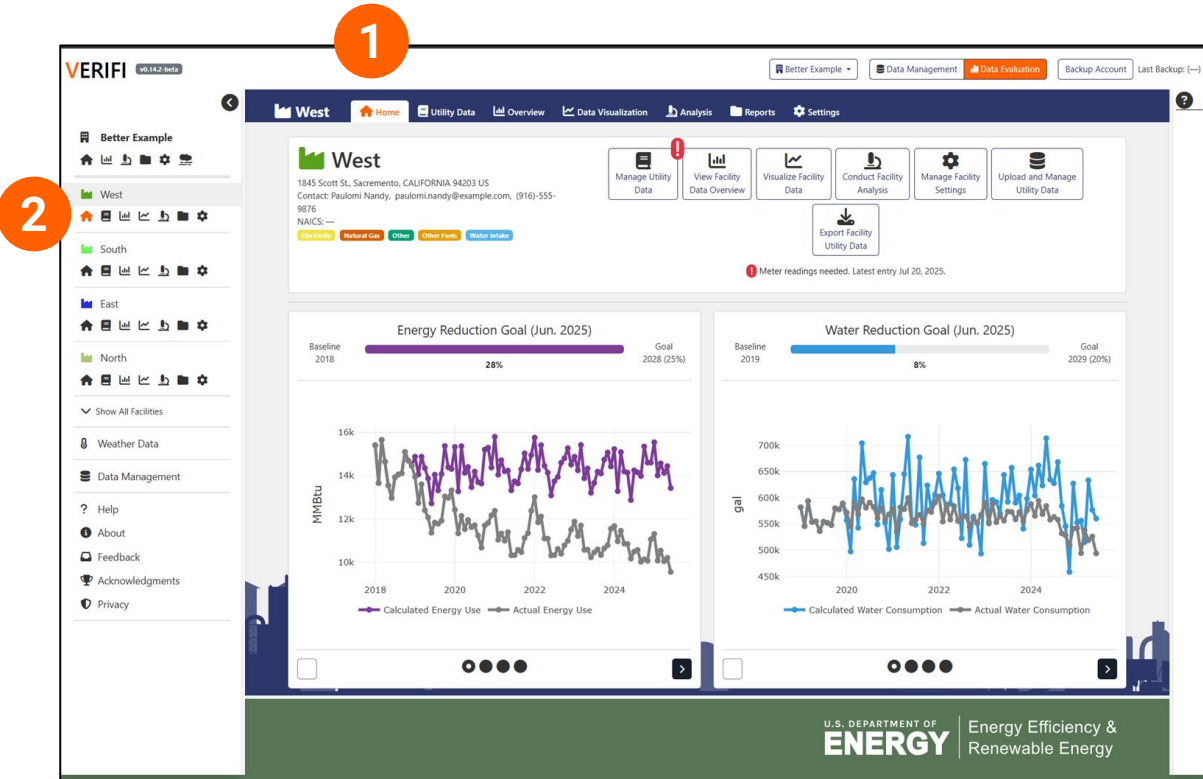
kW

\$

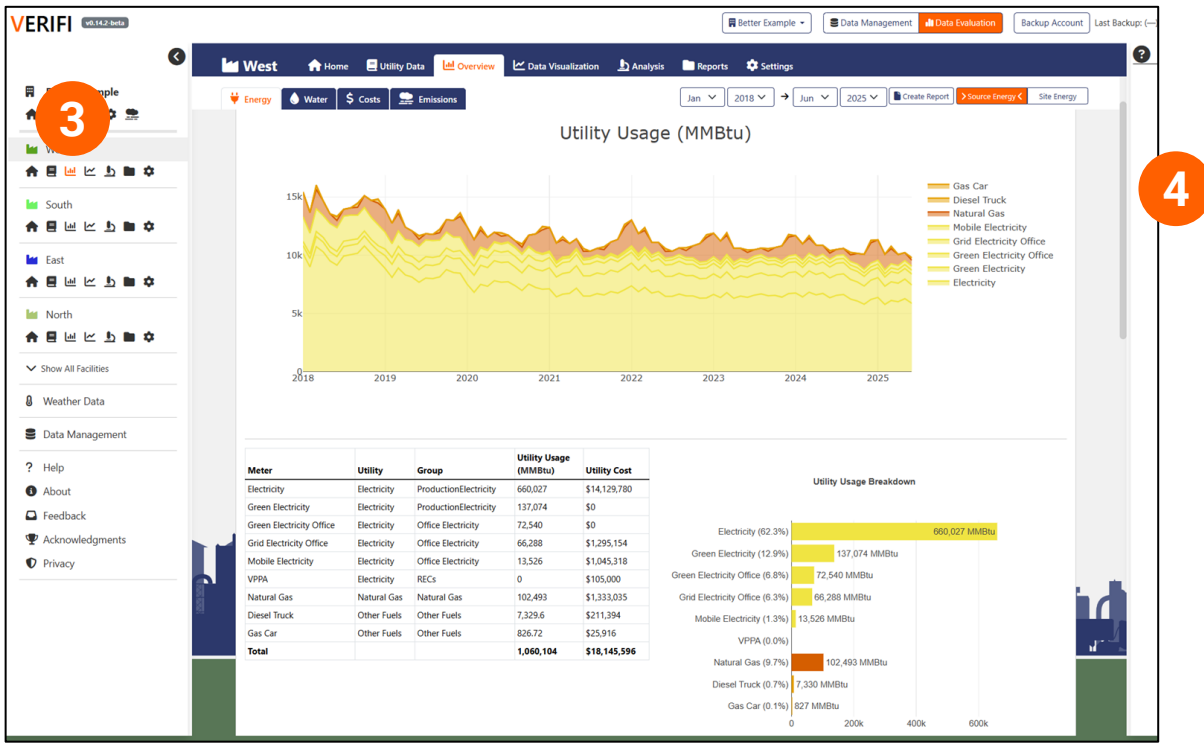


Understand Facility Usage

- Select “Facility Home”
- Explore graphs and usage trends.



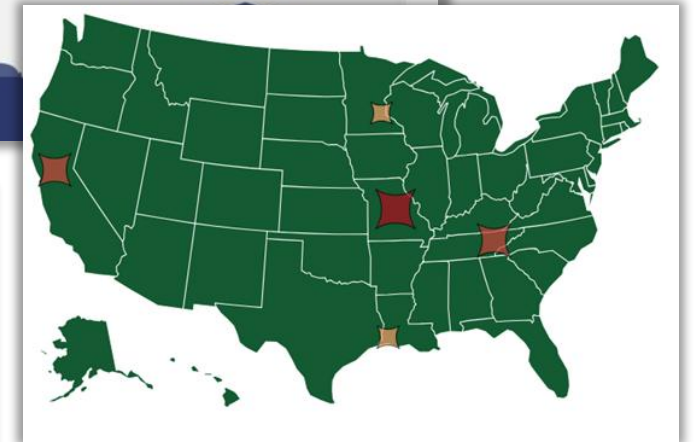
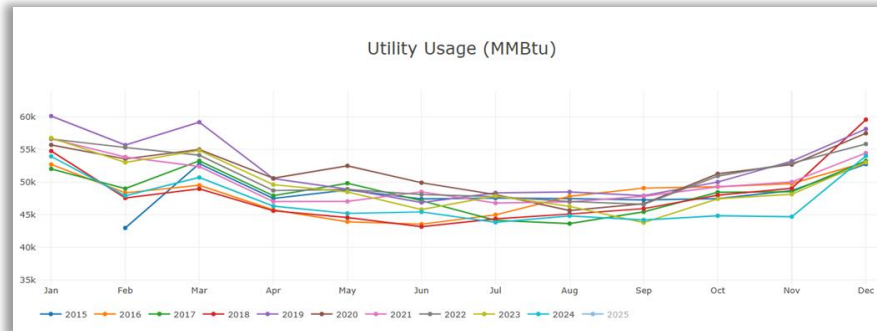
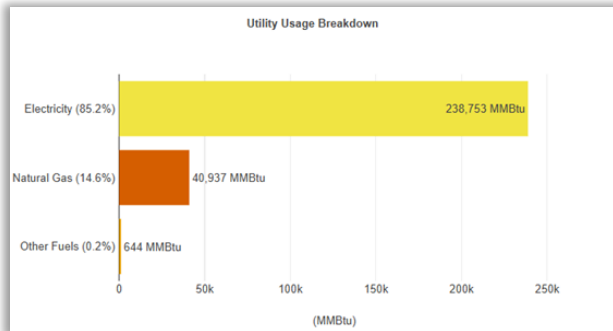
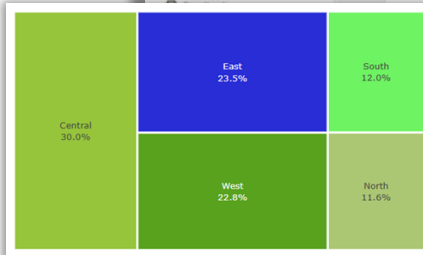
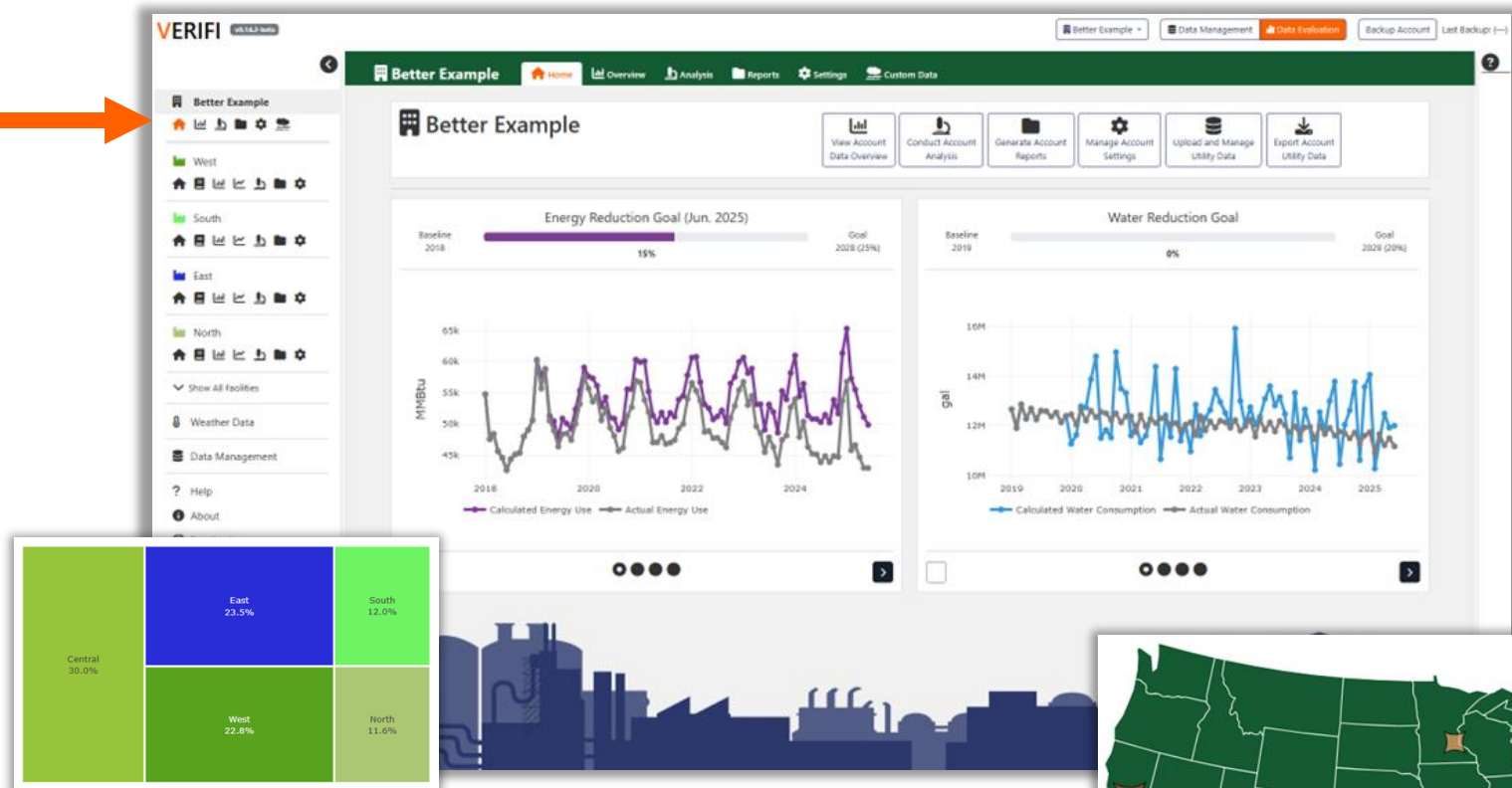
- Select “Facility Overview”
- Explore graphs and usage trends for energy, water, and cost.



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Explore Company Usage

- Select “Company Home” from the side panel.
- Select “Company Overview” from the side panel.
- See breakdowns by facility, utility, and year.
- See data on energy, water, cost, and more.



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Adding Custom Fuels

- VERIFI has in-built databases  for:

- **Scope 1**

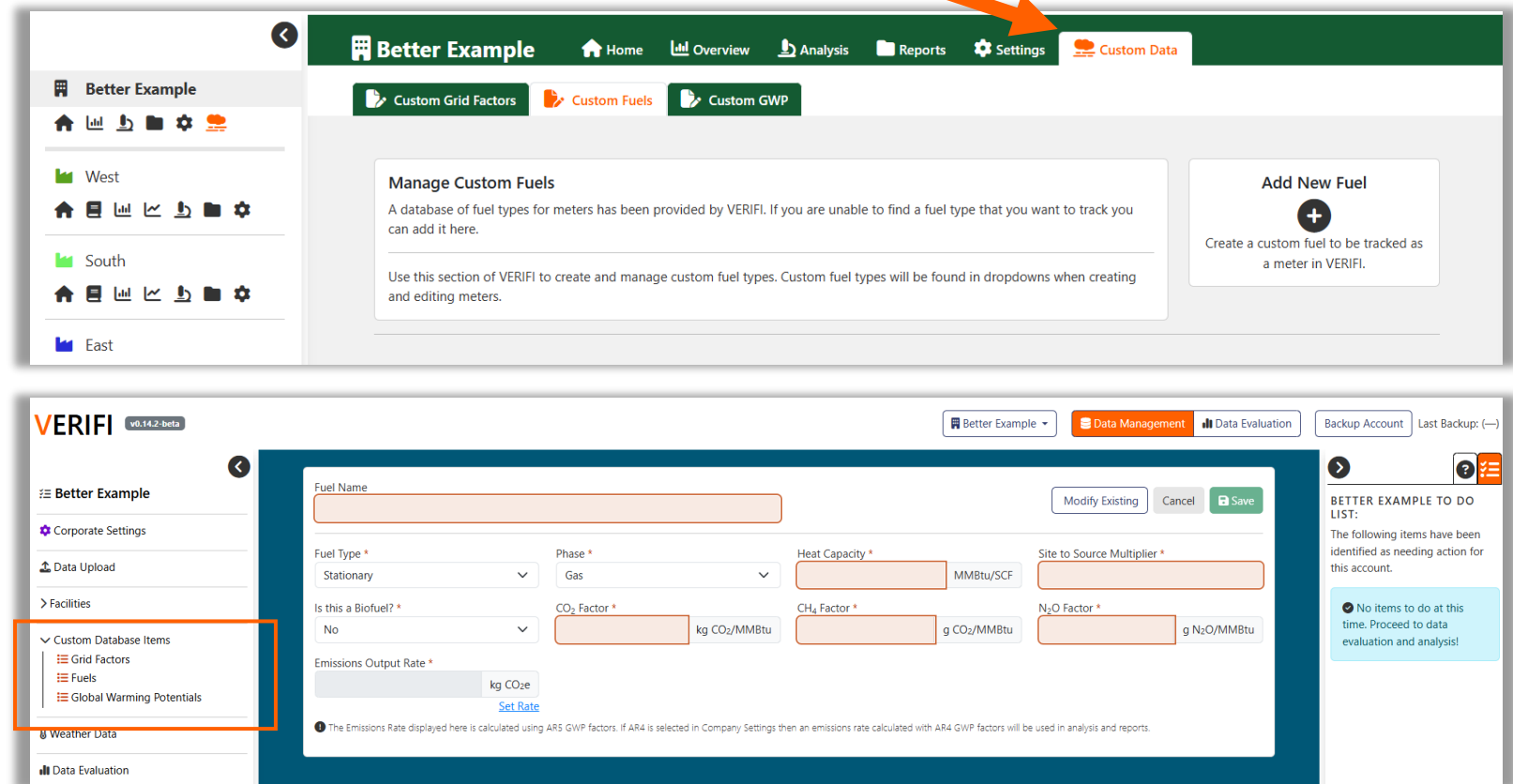
- Stationary (natural gas, propane, fuel oils, etc.)
- Mobile (gasoline, diesel, etc.)
- Fugitive (common refrigerants, etc.)

- **Scope 2**

- Location-based (eGrid)
- Market-based (Green-E)

- **Data Management:** Go to “Custom Database Items”

- You can create factors for custom fuels, process/fugitive releases, and grid factors.
- Data Evaluation: Go to “Custom Data”



The image shows two screenshots of the VERIFI web application. The top screenshot displays the 'Custom Data' section, which includes a sidebar with navigation options like 'Better Example', 'West', 'South', and 'East'. The main content area has tabs for 'Custom Grid Factors', 'Custom Fuels', and 'Custom GWP'. An orange arrow points to the 'Custom Data' tab in the top navigation bar. The bottom screenshot shows the 'Add New Fuel' form, which includes fields for 'Fuel Name', 'Fuel Type' (Stationary), 'Phase' (Gas), 'Heat Capacity' (MMBtu/SCF), 'Site to Source Multiplier', 'Is this a Biofuel?' (No), 'CO₂ Factor' (kg CO₂/MMBtu), 'CH₄ Factor' (g CO₂/MMBtu), 'N₂O Factor' (g N₂O/MMBtu), and 'Emissions Output Rate' (kg CO₂e). A 'Set Rate' link is also present. The sidebar on the left highlights 'Custom Database Items' with a red box, showing sub-items like 'Grid Factors', 'Fuels', and 'Global Warming Potentials'.

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Analysis Overview

Performing analysis and generating reports with VERIFI.



What do I need to perform an analysis?



Before doing an analysis, you must have done the following:

- Entered consumption data
- Added production related predictors (widgets, hours, etc.)

After these initial steps, there are still a few things to do:

- Calendarize your data to fix inconsistent billing periods
- Group meters in similar areas, by process, etc.
- Generate weather predictors
- Explore data relationships between predictors and usage

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Calendarizing Data in VERIFI

- Sometimes billing periods depend on when the utility reads the meter
- *Calendarization* makes billing periods consistent between years
- VERIFI can calendarize for you!

The screenshot shows the VERIFI v0.14.2-beta interface. On the left is a sidebar with navigation options: Better Example, Corporate Settings, Data Upload, Facilities (Manage Facilities, West, South, East, North, Central), Facility Settings, Meters (Manage Meters, Meter Grouping, Electricity), Meter Settings, Meter Readings, Meter Data Quality Re..., Meter Monthly Data, Meter Utility Bill Inspe..., Municipal Water, Natural Gas, RECs, and VPPA. A red arrow points to the 'Meter Monthly Data' option. The main content area displays the 'Electricity Calanderized Data' table. The table has columns: Month, Total Energy (MMBtu), Market-Based Emissions (tonne CO₂e), Total Location-Based Emissions (tonne CO₂e), and Total Cost. The data is for the year 2025, with months from July to June. A green box highlights the table, and a green arrow points from the 'Select Method' dropdown to the 'Calendarization Method' modal. The modal contains a 'Calendarize Meter Data' button and a text box explaining the process: 'Calendarization is the process of allocating data to a consistent and appropriate time period. This is done for reporting purposes so that there can be a fair comparison of utility data for each year and resource. The standard way to calendarize utility data is to divide consumption for each utility by the number of days on the bill. Usage is then allocated to each month based on this daily use ratio and the number of billing days in each month.' A blue arrow points to the 'Calendarize Meter Data' button. Below the modal is a calendar view for December 2019, January 2020, February 2020, and March 2020, showing days of the week and dates. A purple box highlights the calendar view.

Month	Total Energy (MMBtu)	Market-Based Emissions (tonne CO ₂ e)	Total Location-Based Emissions (tonne CO ₂ e)	Total Cost
June, 2025	12,927	807.01	714.33	\$131,738
May, 2025	13,245	826.85	731.89	\$135,420
April, 2025	12,792	798.59	714.33	\$129,986
March, 2025	11,860	827.77	732.7	\$136,170
February, 2025	11,557	730.35	646.48	\$120,259
January, 2025	12,611	787.67	697.21	\$128,606
December, 2024	12,665	790.67	774.45	\$128,752
November, 2024	12,295	767.57	751.83	\$125,342
October, 2024	12,776			
September, 2024	12,540			
August, 2024	12,981			
July, 2024	13,124			

- Navigate to “Monthly Meter Data”
- Click “Select Method”
- Use dropdown to select method used to calendarize the data
- Repeat for all meters that require calendarization

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Grouping Meters for Analysis in VERIFI

- Analysis Groups are collections of meters that are used for analysis
- Group meters in ways that works for your analysis

VERIFI v14.2-beta

Better Example - Data Management Data Evaluation Backup Account Last Backup: (---)

Better Example

Corporate Settings

Data Upload

Facilities

Manage Facilities

West

Facility Settings

Meters

Manage Meters

Meter Grouping

Diesel Truck

Electricity

Gas Car

Green Electricity

Green Electricity Office

Grid Electricity Office

Mobile Electricity

Municipal Water

Natural Gas

Other Water

VPPA

lbProcess

lbRefrigerantA

Predictors

South

East

North

Central

Custom Database Items

Grid Factors

Fuels

Global Warming Potentials

Weather Data

Data Evaluation

Help

About

Feedback

Acknowledgments

Privacy

West Meter Groups

Import Data + Add New Meter Group

Energy Groups

Production Electricity

Meter Source Scope

Electricity Electricity Purchased Electricity

Green Electricity Electricity Purchased Electricity

Natural Gas

Meter Source Scope

Natural Gas Natural Gas Stationary

Other Fuels

Meter Source Scope

Diesel Truck Other Fuels Mobile

Gas Car Other Fuels Mobile

Office Electricity

Meter Source Scope

Green Electricity Office Electricity Purchased Electricity

Grid Electricity Office Electricity Purchased Electricity

Mobile Electricity Electricity Purchased Electricity

Water Groups

Water Intake

Meter Source Scope

Municipal Water Water Intake

Other Water Water Intake

Other Groups

Other

Meter Source Scope

lbProcess Other Process

lbRefrigerantA Other Fugitive

RECs

Meter Source Scope

VPPA Electricity Purchased Electricity

BETTER EXAMPLE TO DO LIST:

The following items have been identified as needing action for this account.

No items to do at this time. Proceed to data evaluation and analysis!

Select "Meter Groupings" from the second level tabs.

Create new or edit existing groups based on meters

Create groups that fit how your energy is used

- You can combine or separate similar utilities (separate production vs office space; combine green meters with grid)
- Group mobile "meters" together to exclude from later energy analyses, but keep for other reports
- Put all other "meters" and RECs in "Other Groups" to separate from energy analyses

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Grouping Meters for Analysis in VERIFI

- Facility energy usage can be affected by the local climate
- VERIFI can automatically generate weather data for your analysis

Weather Data Used by VERIFI

This page provides a detailed look of the weather data available for use by VERIFI. The weather data is pulled from [The National Centers for Environmental Information](#). If quality data is found, VERIFI can automatically calculate heating and cooling degree days, average relative humidity and average dry bulb temperature. The calculated weather data can be used as predictor data for analysis.

For much more information on all things degree days, visit <https://www.degree-days.net/>

Average relative humidity and dry bulb temperature are calculated using a weighted average where the weights are the number of minutes that a reading covers.

Search For Weather Stations

1. Use the location search below to find latitude and longitude associated with a location.
2. Select the location from the table populated on the right.
3. "Fetch Stations" will retrieve the list of closest weather stations in our system and display below.

Location Search

Use the Lat/Long search to find locations.

Search By Facility

Select Location to Search For Stations

Weather stations can be found using geocoordinates.

Type: ☐ City ☐ Display Name (Lat, Long)

City: Oak Ridge, Anderson County, East Tennessee, Tennessee, United States (36.0104, -84.2696)

Furthest Distance: 75 miles

Fetch Stations

Station Name | **State** | **Distance Away** | **Data Start** | **Data End** | **% Hourly Coverage**

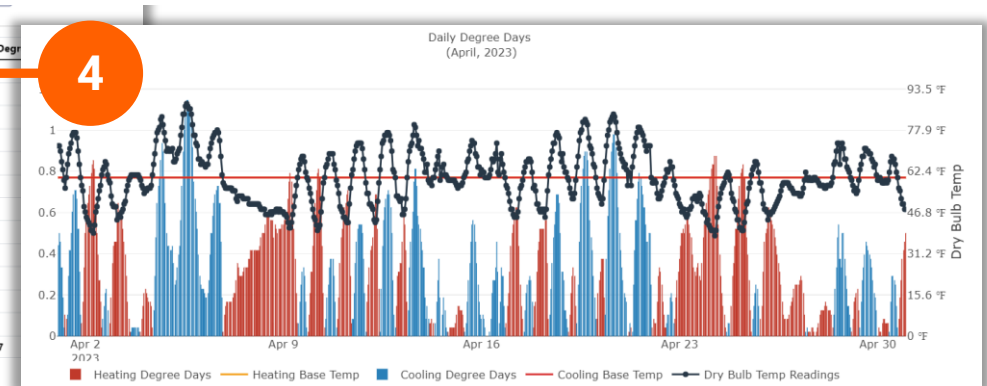
OAK RIDGE (36.023, -84.234)					
ATDL (35.998, -84.219)					
MC GHEE TYSON AIRPORT (35.818, -83.966)					
KNOXVILLE DOWNTOWN ISLAND (35.964, -83.874)					
CAMPBELL CO (36.335, -84.162)					
CROSSVILLE MEMORIAL - WHITSON FIELD APT (35.951, -85.531)					
MIDDLESBORO-BELL COUNTY AIRPORT (36.611, -83.730)					
WILLIAMSBURG (WHITLEY CO APT) (36.8, -84.2)					
ANDREWS-MURPHY AIRPORT (35.195, -83.865)					
WAYNE COUNTY AIRPORT (36.855, -84.856)					
UPPER CUMBERLAND REGIONAL AIRPORT (36.056, -85.531)					
MACON COUNTY AIRPORT (35.223, -83.419)					

Select Year: 2023 | Select Data: Both Degree Days | Heating Base Temperature: 60 °F | Cooling Base Temperature: 60 °F | Create Facility Degree D

Monthly Degree Days (2023)

Month	Heating Degree Days	Cooling Degree Days
Jan. 2023	488.081	9.208
Feb. 2023	299.94	44.958
Mar. 2023	302.836	64.208
Apr. 2023	128.667	123.233
May. 2023	34.254	248.438
Jun. 2023	2.771	374.31
Jul. 2023	0	548.65
Aug. 2023	0	527.562
Sep. 2023	3.417	366.982
Oct. 2023	109.458	155.006
Nov. 2023	298.23	53.354
Dec. 2023	476.956	1.687
Total	2,144.61	2,517.597

1. Select "Weather Data" from side bar
2. Enter ZIP code / city or select a facility and set a distance radius for weather stations
3. Select a station to view detailed weather data
4. Set balance temperatures, view data integrity, and assign data to a facility
5. Update quickly with "Bulk Update"



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Visualizing Data in VERIFI

- Use data visualizations to: 
- Identify trends
- Find correlations to predictors
- Prepare for regression analysis
- View data by:
 - Time series
 - R2 variance
 - Data correlation



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What kinds of analysis and reports can VERIFI do?

- Reports are aggregates of data or analysis results for viewing in a concise matter
- Some VERIFI reports...
- Match formal reporting templates
- Have been developed to help compare facility-level performance
- If you have ideas on reports that can be integrated into VERIFI let us know!

Types of Company-level Reports

- Better Plants Energy & Water
- Energy Performance
- Data Overview

Types of Facility-level Reports

- Analysis (Results & Regression Quality)
- Data Overview

The screenshot shows a 'Better Plants' Annual Reporting Form. The header includes the 'Better Plants' logo and the text 'U.S. DEPARTMENT OF ENERGY'. The form is titled 'Annual Reporting Form' and 'BETTER BUILDINGS, BETTER PLANTS'. It is generated by VERIFI. The form includes a 'Contact Information' section with fields for Name, Phone, and Email. Below this is a table with three columns: 'Baseline Year 2018', 'Report Year 2023', and an unlabeled column for the metric. The table contains data for 'Number of Participating Plants', 'Primary Energy Consumed (MMBtu)' (broken down by Electricity and Natural Gas), and 'Total Primary Energy Consumed (MMBtu)'.

	Baseline Year 2018	Report Year 2023
Number of Participating Plants*	3	4
Primary Energy Consumed (MMBtu)		
Electricity	347,464	355,594
Natural Gas	45,197	69,043
Total Primary Energy Consumed, (MMBtu)	392,661	424,637

DRAFT

What are facility-level analyses?

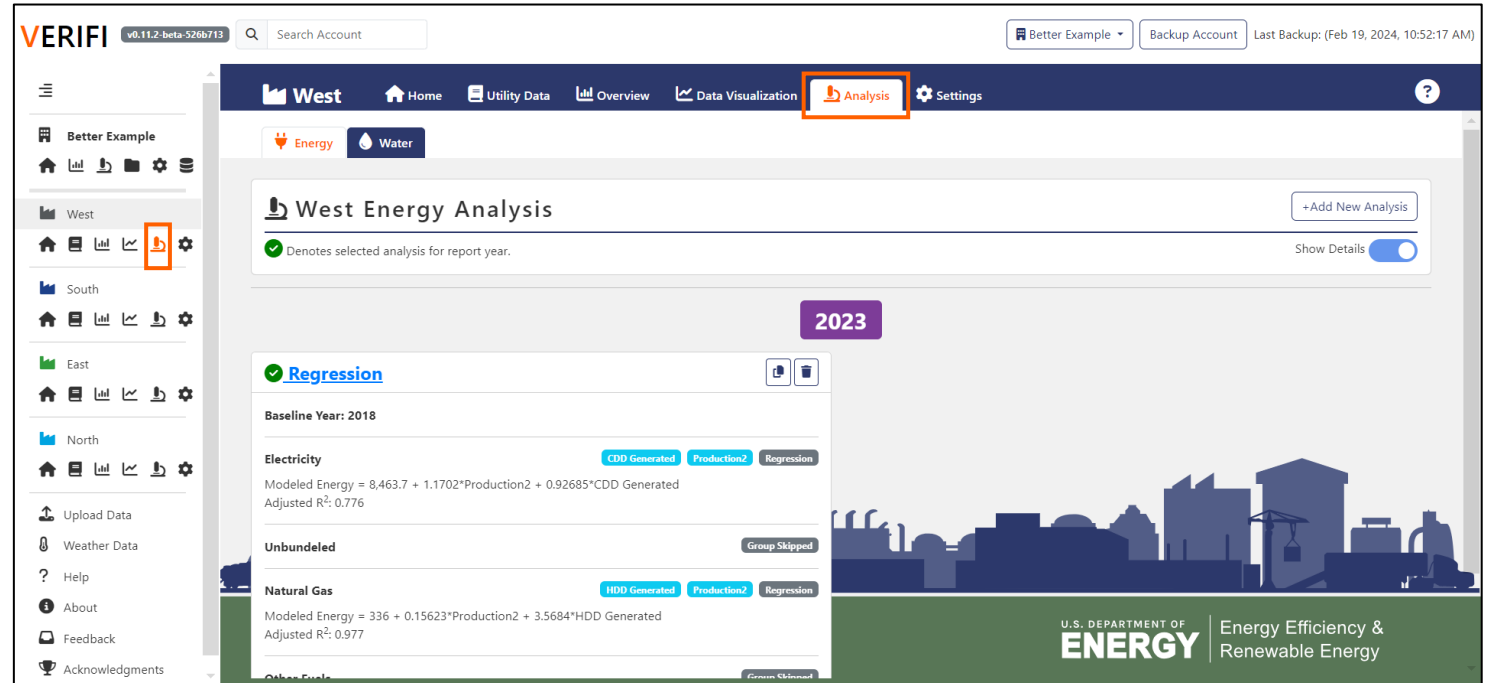
Facility-level analysis tools help you to:



- Understand how energy and water efficient your facilities are
- Normalize usage data

VERIFI can perform many types of analysis including:

- Absolute
- Intensity
- Regression



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Completing a Facility-Level Analysis

- 1. Name your analysis and select the baseline year
- 2. Select the analysis type (e.g., absolute, intensity, regression)
- 3. IF A REGRESSION – Pick the best model for the facility analysis
- 4. Repeat for all groups!

1

West

Home

Utility Data

Overview

Data Visualization

Analysis

Settings

Better Example

West

South

East

North

Upload Data

Weather Data

Help

About

Feedback

Acknowledgments

Analysis Setup

Electricity

Unbundled

Natural Gas

Other Fuels

Facility Summary

Account Analysis

All Facility Analysis

Analysis Setup

This analysis is used in one or more account analysis. Click the "Account Analysis" tab to view and navigate to the corresponding analysis items. Making changes to this analysis may effect the results of the account analysis.

Name

Regression

Baseline Year

2018

Energy displayed as site or source?

Source Energy

Report Year

2023

Energy Unit

Million British Thermal Units (MMBtu)

Reporting Year

Calendar Year

Fields set in "Settings" tab.

3

Analysis Options

Regression Model

Annual Analysis

Monthly Analysis

Select Model	Model Year	Variable p-Values	R2	Adjusted R2	Model p-Value	Formula	Model Notes
☒	2020	Production2: 0.00 CDD Generated: 0.01	0.817	0.776	0.00	8,460 +(1.17*Production2) +(0.927*CDD Generated)	
☐	2020	Production2: 0.00 HDD Generated: 0.05	0.742	0.685	0.00	8,940 +(1.09*Production2) +(-1.00*HDD Generated)	HDD Generated coef < 0
☐	2018	Production1: 0.05 Production2: 0.01	0.703	0.637	0.00	8,150 +(0.0778*Production1) +(1.44*Production2)	
☐	2020	Production2: 0.00	0.602	0.562	0.00	8,590 +(1.22*Production2)	
☐	2020	Production1: 0.00	0.591	0.550	0.00	5,690 +(0.142*Production1)	
☐	2018	Production2: 0.01	0.531	0.484	0.01	10,200 +(1.77*Production2)	Adjusted R2 < .5
☐	2021	Production1: 0.02 HDD Generated: 0.18 CDD Generated: 0.16	0.578	0.420	0.06	7,080 +(0.0738*Production1) +(1.52*HDD Generated) +(1.33*CDD Generated)	Adjusted R2 < .5

2

Analysis Setup

Electricity

Unbundled

Natural Gas

Other Fuels

Facility Summary

Account Analysis

Analysis Options

Regression Model

Annual Analysis

Monthly Analysis

Electricity Setup

This analysis is used in one or more account analysis. Click the "Account Analysis" tab to view and navigate to the corresponding analysis items. Making changes of the account analysis.

1. Select Analysis Type

Absolute

Classic Intensity

Regression

Exclude from Analysis

3. Add Baseline Adjustment

2. Add Data Adjustment

4. Regression Model

Use "Regression Model" tab to setup the regression model for this group.

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Making Facility-Level Reports

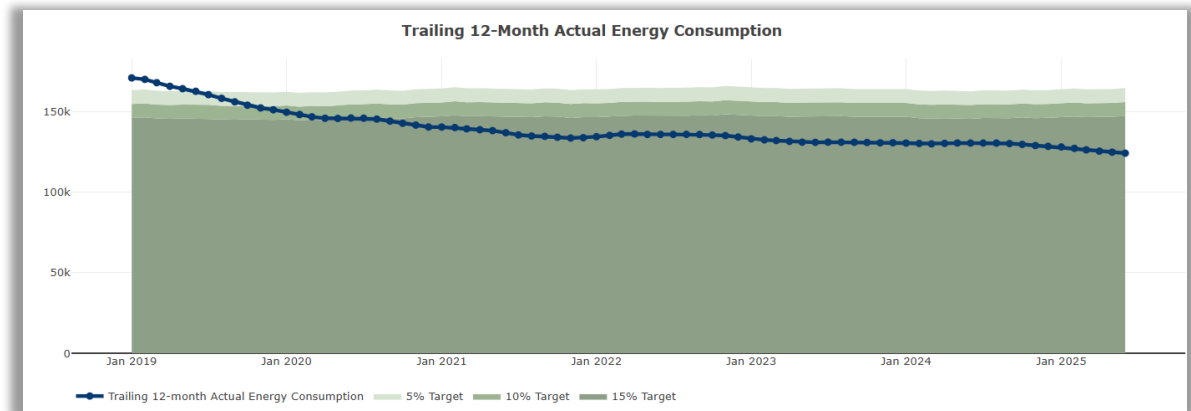
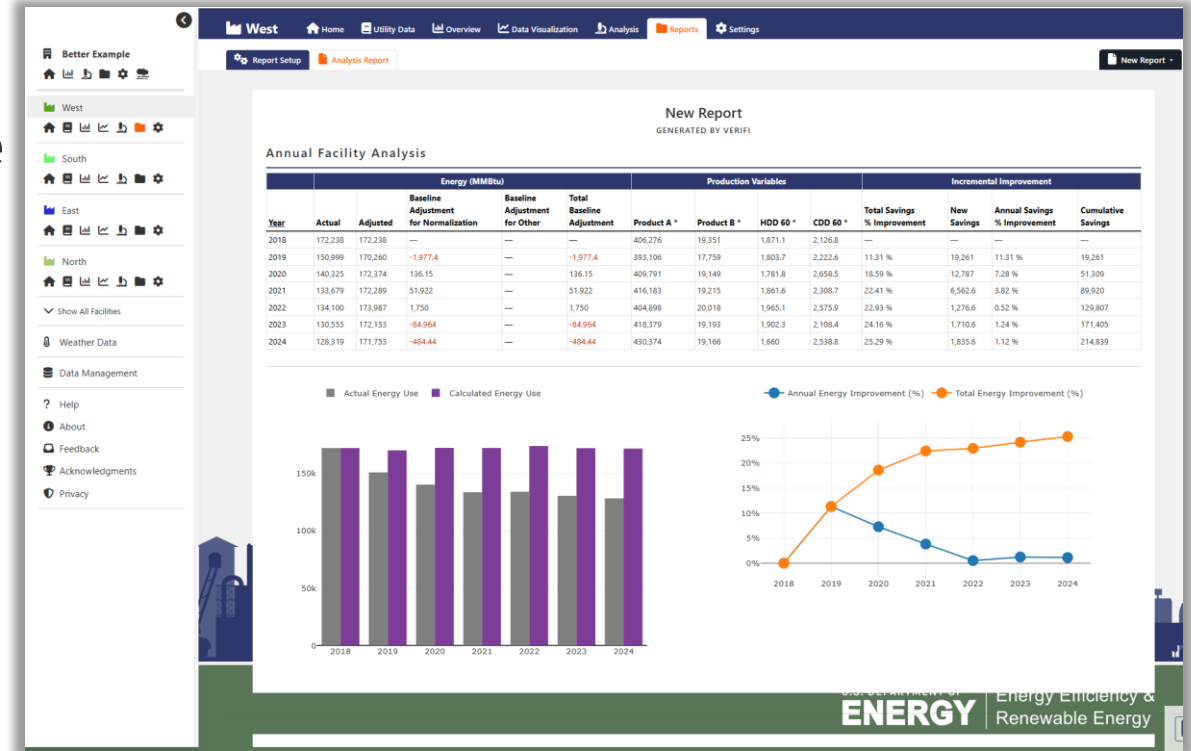
Facility reports provide a quick overview of the analyses and data.



VERIFI can make:

- Analysis Report
 - Analysis Results
 - Regression Statistics (selected variables, R2, p-value, validation checks)
- Data Overview Reports
- Savings Reports
 - Savings as up-to-date as your data entry

Select a facility-level analysis to make an automated report



DRAFT

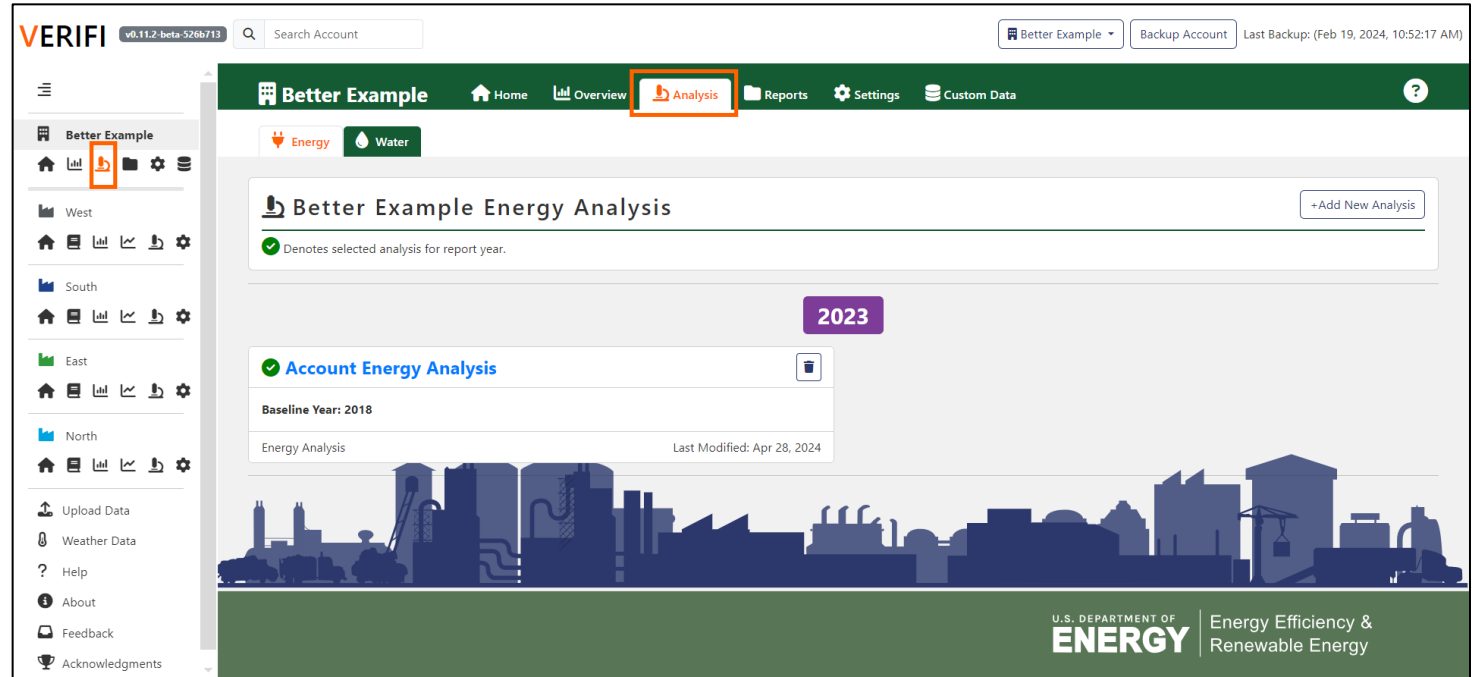
What are company-level analyses?

Roll-up Analysis: 

- Necessary for Better Plants energy and water reports
- Choose which facility-level analyses to include
- Calculate Company-level performance

Facility-level reports must have the same*:

- Baseline year
- Report year
- * Except for new facilities marked in Settings



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Completing a Company-Level Analysis

- 1. Start a Company-level analysis
- 2. Set baseline and reporting years for the analysis
- 3. Select which facility-level analyses to include
- 4. View the results!

VERIFI v0.11.2-beta-520b713

Better Example Home Overview Analysis Reports Settings Custom Data

Better Example Energy Analysis

2023

Account Energy Analysis

Baseline Year: 2018

Energy Analysis

Last Modified: Apr 28, 2024

Analysis Setup

This analysis is used in one or more reports. Changes to this analysis may effect the results of the corresponding reports.

Name: Account Energy Analysis

Baseline Year: 2018

Energy displayed as site or source?: Source Energy

Report Year: 2023

Energy Unit: Million British Thermal Units (MMBtu)

Reporting Year: Calendar Year

Fields set in Settings tab.

This analysis is used in one or more reports. Changes to this analysis may effect the results of the corresponding reports.

Analysis Items

+Add Facility Analysis

Analysis Item	Group Analysis	Last Modified	
☒ Regression	Electricity: Regression Unbundled: Exclude from Analysis Natural Gas: Regression Other Fuels: Exclude from Analysis	4/28/24, 10:07 PM	View Analysis
☐ Skip Facility			

Annual Analysis Monthly Analysis Facility Summaries

Account Analysis

+Create BP Report

Choose Columns

Table Graph

Year	Energy (MMBtu)				Incremental Improvement						
	Actual	Adjusted	Baseline Adjustment for Normalization	Baseline Adjustment for Other	Total Baseline Adjustment	SEnPI	Savings	Total Savings % Improvement	New Savings	Annual Savings % Improvement	Cumulative Savings
2018	392,661	392,661	—	—	—	1	—	—	—	—	—
2019	451,253	468,967	-2,682.4	78,988	76,306	0.96223	17,714	3.78 %	17,714	3.78 %	17,714
2020	439,999	473,434	1,785	78,988	80,773	0.92938	33,435	7.06 %	15,721	3.29 %	51,149
2021	423,010	468,386	-3,262.7	78,988	75,725	0.90312	45,376	9.69 %	11,941	2.63 %	86,525
2022	436,680	483,467	11,818	78,988	90,806	0.90323	46,786	9.68 %	1,410.3	-0.01 %	143,311
2023	424,637	472,531	881.62	78,988	79,870	0.89864	47,894	10.14 %	1,107.7	0.46 %	191,205



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Making Company-Level Reports

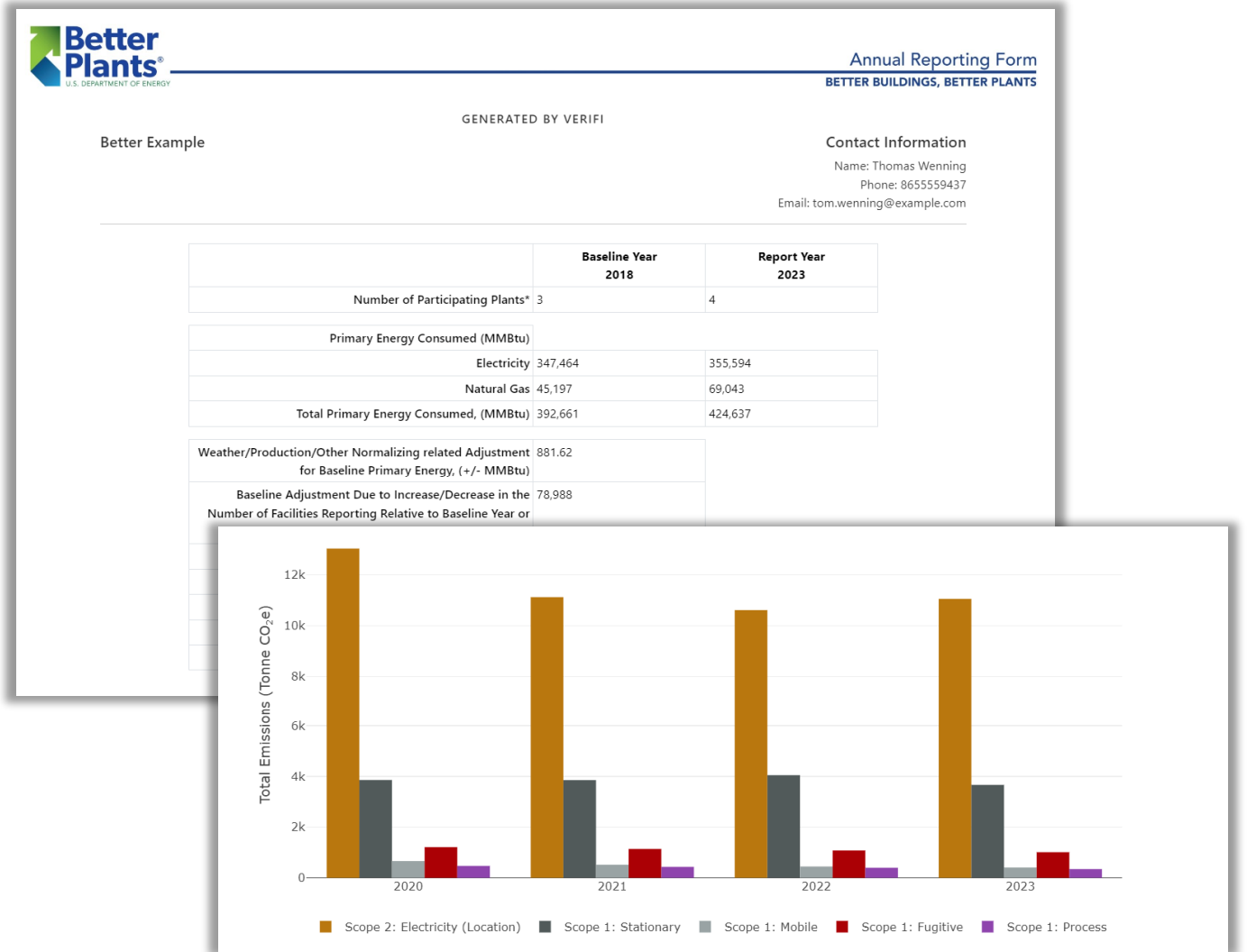
Reports are aggregates of analysis results for concise viewing.



VERIFI can make:

- Better Plants Reports
- Better Climate Challenge Reports
- Better Plants Water Reports
- Energy Performance Reports
- Data Overview Reports
- Savings Report
- Modeling Report

Select a company-level analysis to make an automated report.



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Need more help?

Visit the [Industrial Resources VERIFI page](#) for:

- This guide
- Tips & Tricks
- FAQ covering data entry to reports

VERIFI Tips & Tricks

Just wanting Intensity or Absolute results? Bulk Analysis

1. Set up a **Company-level Analysis** (set year, give it a good name)
2. Click "Create Bulk Analysis"
3. Select Absolute or Classic Intensity
4. Review each facility, View the facility analysis to modify
5. Return quickly under "Account Analysis"

The image shows a screenshot of the VERIFI software interface with numbered callouts (1-5) highlighting the steps for Bulk Analysis. The interface includes a sidebar with navigation options like 'About VERIFI', 'Software Downloads', 'User Guides', 'Case Studies', and 'Citations'. The main content area displays the 'Bulk Analysis' guide with a 'DRAFT' label at the bottom.

VERIFI FAQ

Which data input method should I use?

```
graph TD
    Q1[Do you have a VERIFI Export file?] -- Yes! --> A1[VERIFI Export file]
    Q1 -- Nope --> Q2[Are you entering first time or updating with new data?]
    Q2 -- Updating --> Q3[How many sites do you want to enter data for?]
    Q2 -- It's my first time --> Q4[How many sites do you want to enter data for?]
    Q3 -- I think it's a lot... --> A2[Template]
    Q3 -- Not too many --> A3[Manual Entry]
    Q4 -- I think it's a lot... --> A4[Template]
    Q4 -- Not too many --> Q5[Do you want to enter anything more than energy data?]
    Q5 -- Nope --> A5["EnPI Format"]
    Q5 -- Yes! --> A6[Template]
```

The image shows a screenshot of the VERIFI FAQ page with a decision tree for data input methods. The tree starts with the question 'Which data input method should I use?' and branches based on whether the user has a VERIFI Export file, if it's their first time or updating, and how many sites they want to enter data for. The final options include 'VERIFI Export file', 'Template', 'Manual Entry', 'EnPI Format', and 'Template'. The page also includes a 'DRAFT' label and a 'How will I enter next year's data?' section with the option '1. Download data into template'.

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Feedback or bugs?

Email:

verifi-help@ornl.gov

armstrongko@ornl.gov

accawigk@ornl.gov

wenningtj@ornl.gov



Other DOE Software Tools

Helpful software tools developed for the industrial sector.



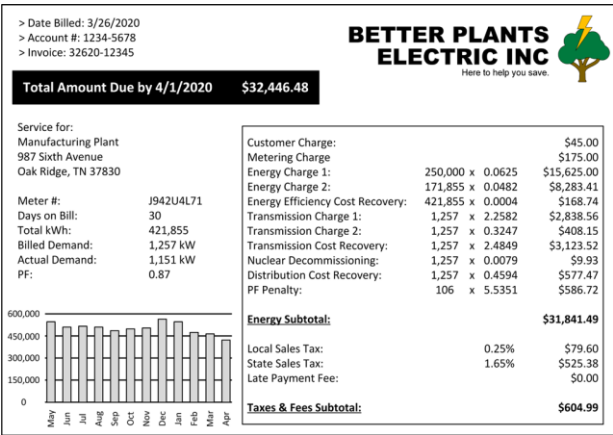
50001 Ready Navigator

Facility personnel don't always have the time, resources, or knowledge to build a near ISO compliant energy management system (EnMS), especially without customers driving it. Developing an EnMS with the Navigator can reduce operational costs and uncertainty, help quality for rebates, and influence your value chain.

50001 Ready Navigator helps manufacturers build a near ISO compliant energy management system, **by themselves.**

Context

Who all is involved?
What is the scope?



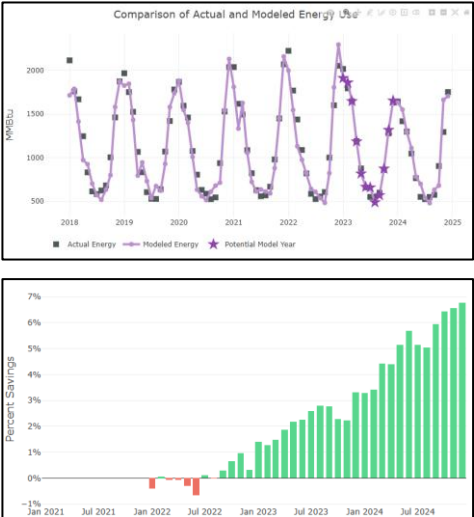
Understand

How do we collect, store, and analyze data?
What are my main energy users?



Continuous Improvement

What are our goals and how will we get there?
Training, Awareness, Documentation



	Baseline Year 2021	Report Year 2024
Number of Participating Plants*	5	5
Primary Energy Consumed (MMBtu)		
Electricity	506,192	483,553
Natural Gas	92,749	80,973
Total Primary Energy Consumed, (MMBtu)	598,942	564,526
Weather/Production/Other	6,579.9	
Normalizing related Adjustment for Baseline Primary Energy, (+/- MMBtu)		
Baseline Adjustment Due to Increase/Decrease in the Number of Facilities Reporting Relative to Baseline Year or Other Operational Changes, (+/- MMBtu)	74,702	
Adjusted Baseline of Primary Energy (MMBtu)	680,223	
New Energy Savings for Current Year (MMBtu)	20,593	
Unbanked Energy Savings (MMBtu)	115,697	
Total Energy Savings since Baseline Year (MMBtu)	115,697	
Annual Improvement In Energy Intensity for Current Year (%)**	3.45 %	
Total Improvement In Energy Intensity (%)**	17.01 %	

DRAFT

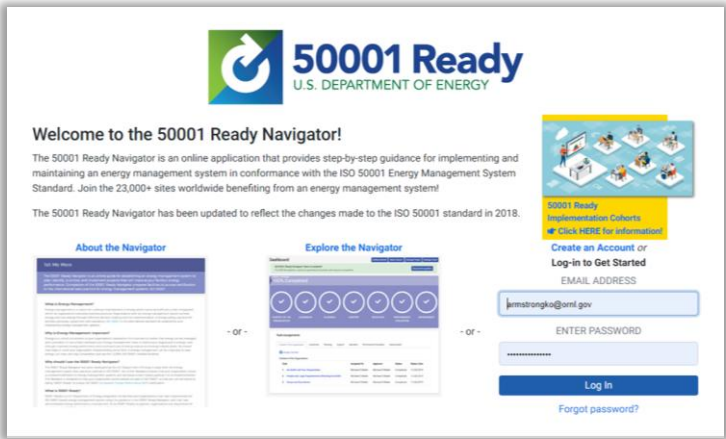
50001 Ready Navigator

Helps energy and sustainability teams create a near-ISO 50001 compliant energy management system. **Key Features:**

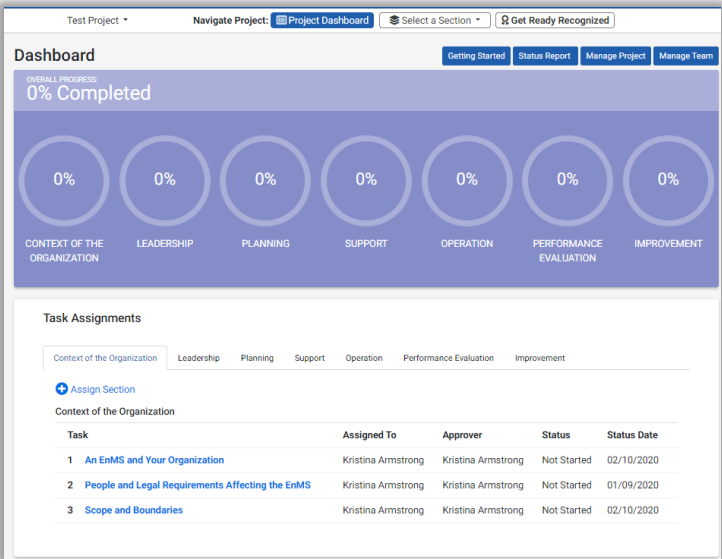
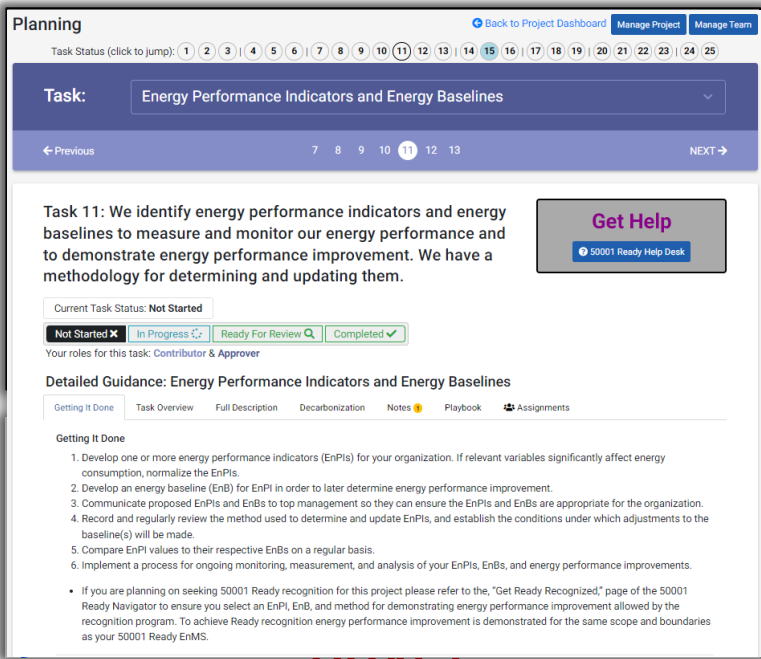
- Step-by-step “Turbo Tax” approach to 50001
- 25 tasks divided into 7 sections
- Extensive guidance available in each module
- Self-attest to completion of 50001 Navigator

Find this tool:

<https://navigator.lbl.gov/>
<https://betterbuildingsolutioncenter.energy.gov/iso-50001/50001Ready>



Cohorts available to help develop EnMS!



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MEASUR - The Energy Assessment Tool

Facility personnel don't always have the **time, resources, or knowledge** to find or assess energy projects. Consultants can be **expensive**, and vendors provide a **"black box"** solution trying to sell their product.

MEASUR helps manufacturing diagnose and triage their systems and understand the potential impact of energy efficiency projects **by themselves**.

Triage

How do you find projects?



Measure

What do you do with these numbers?



Calculate

How does this become energy savings?



DRAFT

MEASUR – Assess equipment's energy efficiency

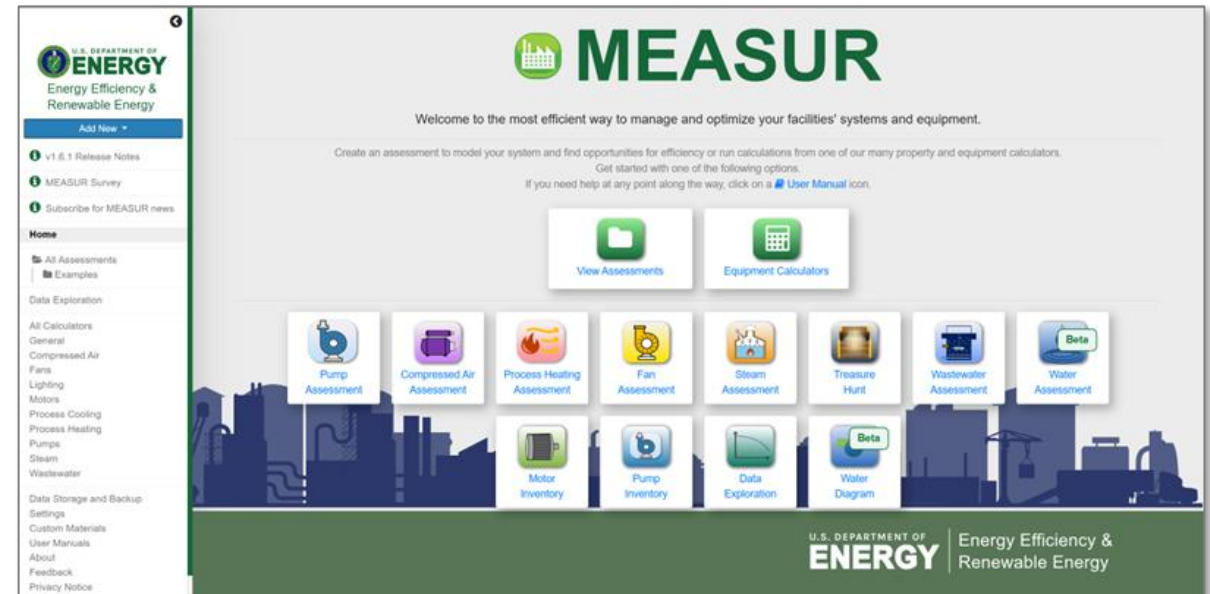
Helps identify, assess, and quantify energy savings opportunities in operations and energy support systems.

Key Features:

- Similar workflow for all assessment modules
- Graphs, figures, accessible help text & examples
- Regular updates

Additional Features:

- FREE & Open source, Cross-platform tool
- Private Data & Global Useability
- Technology and vendor agnostic

The image is a screenshot of a report titled 'Process Heating - Fuel Example' from the MEASUR application. It shows a detailed breakdown of energy losses and opportunities for improvement. The report is organized into columns for different assessment scenarios and a final 'All Opportunities' column. The data is presented in a table format with various energy loss categories and their corresponding values in Btu/hr.

Energy Loss/Use	Baseline	Individual Opportunity 4 - Reduce Oil level in fuel gases	Individual Opportunity 3 - Repair wall insulation	Individual Opportunity 2 - Repair fixed openings	Individual Opportunity 1 - Preheat Charge Material	All Opportunities
Charge Materials	143.96	143.96	143.96	143.96	143.96	143.96
Fixtures, trays etc.	---	---	---	---	---	---
Wall Losses	7.47	7.47	5.90	7.47	7.47	5.90
Cooling Losses	24.16	24.16	24.16	24.16	24.16	24.16
Atmosphere Losses	---	---	---	---	---	---
Opening Losses	2.81	2.81	2.81	1.86	2.81	1.86
Leakage Losses	3.26	3.26	3.26	3.26	3.26	1.83
Extended Surface Losses	---	---	---	---	---	---
Other Losses	---	---	---	---	---	---
Total heat heat Required	181.27	181.27	177.29	180.02	180.91	179.06
Available Heat (%)	89.2%	84.0%	89.2%	89.2%	89.2%	84.0%
Free Gas Losses	124.82	102.11	122.09	123.96	114.25	89.60
Exothermic Heat from Process	---	---	---	---	---	---
Gross Heat Input	206.09	283.38	299.58	300.99	290.16	248.66

Find this tool at
<https://measur.ornl.gov>

DRAFT

JUSTIFI - The Project Valuation Tool

Facility personnel don't always have the **time, resources, or knowledge** to find or assess the true impact of energy projects.

JUSTIFI helps manufacturing energy managers connect with others in their organization to understand the impact of their project on company KPIs, finding the true impact of the project, showing lower ROI and helping push project implementation, **by themselves.**

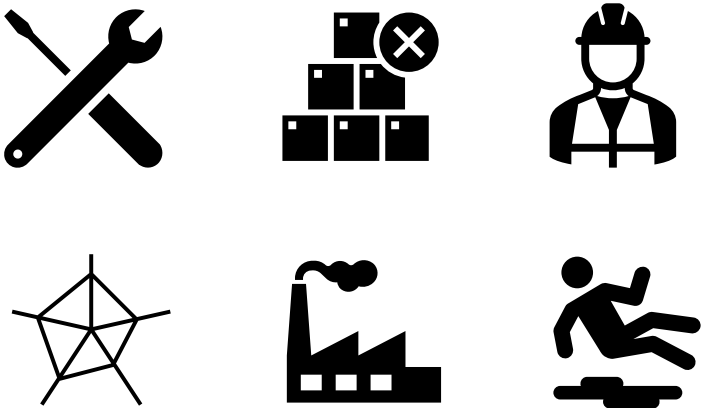
Define

How is this impacting the facility?



Measure

How do you get these numbers?



Assess

How does this become cost savings?

	Without NEBs			With NEBs		
	Implementation Cost (\$)	Financial Impact (\$/yr)	Simple Payback (yrs)	Implementation Cost (\$)	Financial Impact (\$/yr)	Simple Payback (yrs)
1. Insulate Melted Chocolate Tank	\$50,000	\$28,000	1.79	\$50,000	\$85,000	0.59
2. Barring Room Lights	\$15,000	\$1,500	10	\$15,000	\$63,460	0.24
3. Steam Leaks at HP & LP	\$2,500	\$13,080	0.19	\$2,500	\$13,080	0.19
Additional Projects/NEBs	\$50,000	\$8,040	6.22	\$48,500	\$11,890	4.08
Total	\$117,500	\$50,620	2.32	\$116,000	\$173,430	0.67



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JUSTIFI – Understand additional impacts

Helps identify, assess, and quantify the multiple benefits of energy efficiency opportunities to improve financial impact estimations of projects.

Key Features:

- Identify stakeholders, KPIs/KPMs, Equipment, End Uses
- Use to find and quantify NEBs
- Graphs & Figures!
- Accessible help text and examples

Additional Features:

- FREE & Open source, Cross-platform tool
- Private Data & Global Useability



Find this tool at
<https://justifi.ornl.gov/>

DRAFT



Questions?

